SEER Cancer data Analysis

Team name

Cirrus

Presented by

Muhtasim Zaowad

4th year

IASDS

Content

Introduction

EDA

Data Cleaning

Feature Engineering

Model Building

Model Comparison

Conclusion

1. *Introduction*

What is SEER cancer data?

Surveillance, **Epidemiology**, and **End Results** Program

"SEER is the National Cancer Institute's premier **cancer surveillance database**, capturing detailed tumor characteristics, treatments, and long-term survival outcomes for nearly half of the U.S. population."

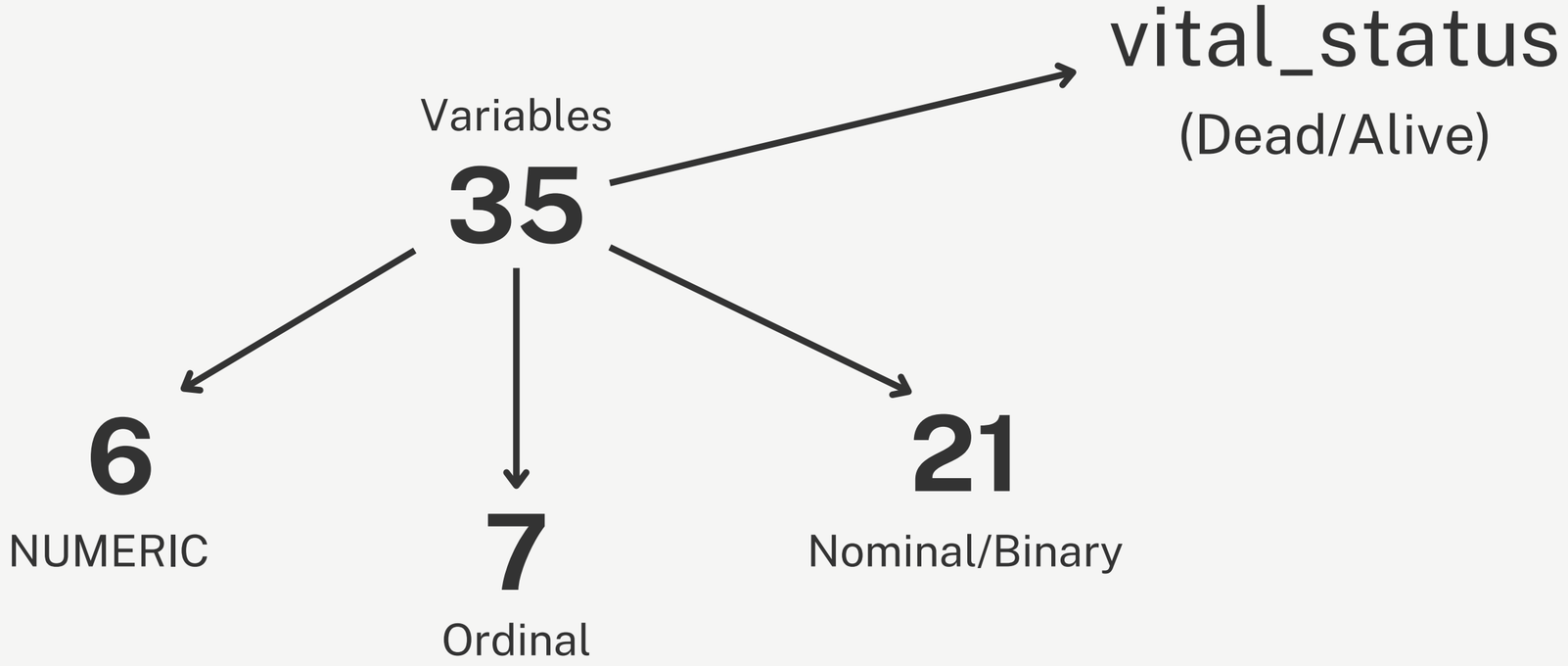
Brief data description

26,000 → training data rows

36,000 → test data rows

35 → variables in total

Brief data description



Variable Description

Variable Groups	Numbers
1. Patient Demographics & Background	[6]
2. Diagnostic Timeline & History	[3]
3. Tumor Anatomy & Histology	[5]
4. Tumor Size Measurements	[3]
5. Cancer Staging & Tissue Extension	[5]
6. Lymph Node Spread	[2]
7. Metastasis Sites at Diagnosis (2010+)	[4]
8. Treatment & Surgical History	[7]

1.Patient Demographics & Background

variable name	dtype
patient_id	identifier
age_recode	ordinal
sex	nominal
race	nominal
origin	nominal
marital_status_at_diagnosis	ordinal

2.Diagnostic Timeline & History

variable name	dtype
year_of_diagnosis	Ordinal
sequence_number	Ordinal
diagnostic_confirmation	Categorical

3.Tumor Anatomy & Histology

variable name	dtype
primary_site	categorical
site_recode_icdo3_who2008	categorical
laterality	categorical
histologic_type_icdo3	categorical
grade_recode_thru2017	ordinal

4.Tumor Size Measurements

variable name	dtype
tumor_size_overtime	numerical
tumor_size_summary	numerical
cs_tumor_size20042015	numerical

5.Cancer Staging and Tissue Extension

variable name	dtype
summary_stage	ordinal
cs_extension20042015	ordinal
derived_eod2018t_recode2018	ordinal
derived_eod2018n_recode2018	ordinal
derived_eod2018m_recode2018	ordinal

6.Lymph Node Spread

variable name	dtype
regional_nodes_examined	numeric
regional_nodes_positive	numeric

7.Metastasis Site At Diagnosis (2010+)

variable name	dtype
seer_combined_metsatdxbone2010	binary
seer_combined_metsatdxbone2010	binary
seer_combined_metsatdxbone2010	binary
seer_combined_metsatdxbone2010	binary

8.Treatment and Surgical History

variable name	dtype
rx_summ_surgprim_site19982022	categorical
rx_summ_surgprim_site2023	categorical
rx_summ_scope_regInsur2003	categorical
rx_summ_surgothregdis2003	categorical
rx_summ_surgradseq	categorical
radiation_recode	categorical
reason_nocancer_directed_surgery	categorical

Why is this not a Lifetime data?

Point 1: **Missing Duration**

- The core issue identified here is that the dataset only has a cross-sectional binary label (vital_status: Alive vs. Dead) with no duration variable at all (no survival_months, no time of event, and no time of censoring). Without an elapsed time, it is just standard binary classification data, not lifetime data.

Point 2: **Temporal Exposure Bias**

- Because the time metric is absent and follow-up horizons vary across calendar years (2013 vs. 2023), treating vital_status as a simple target creates confounding. A 2013 patient who is labeled "Dead" had a decade to experience the event, whereas a 2023 patient labeled "Alive" might die next month.

Structural Change of SEER

Change 1:

- Discontinued Collaborative Stage (CS) schema at the end of 2015 and introduced a new summary tumor size standard in 2016.

Change 2:

- Retired the legacy macro-stage system (summary_stage) and adopted the Extent of Disease (EOD 2018) 8th Edition TNM schema.

Change 3:

- Retired the 4-tier histologic grading code (grade_recode_thru2017) at the end of 2017 to split grade into separate clinical, pathological, and post-neoadjuvant standards.

Change 4:

- Abandoned its legacy 2-digit surgery codes (1998–2022) to align with the ACS Commission on Cancer (CoC) STORE alphanumeric standard (e.g., 'A330' for lobectomy)

Change 5:

- Solid-tumor TNM metastasis staging is structurally inapplicable to non-carcinoma hematologic malignancies (such as primary pulmonary lymphomas and multiple myelomas).

Structural Change 1:

- **On December 31, 2015, NCI retired the Collaborative Stage (CS) standard.**

year_of_diagnosis	**cs_tumor_size20042015**	**tumor_size_summary**
2013	populated	
2014	populated	
2015	populated	
2016		populated
2017		populated
...		

Structural Change 2:

On January 1, 2018, SEER retired its legacy macro-staging system (summary_stage: Localized, Regional, Distant). In its place, SEER adopted the modern Extent of Disease (EOD 2018) system, capturing granular AJCC 8th Edition TNM categories (T1a–T4, N0–N3, M0–M1c).

year_of_diagnosis	**summary_stage**	**derived_eod2018 (T/N/M)**
2013	populated	
...	populated	
2017	populated	
2018		populated
2019		populated
...		...

Structural Change 3:

- **On December 31, 2017, SEER officially retired the historical unified 4-tier grading code (grade_recode_thru2017).**

year_of_diagnosis	**grade_recode_thru2017**
2013	populated
...	populated
2017	populated
2018	
2019	
...	

Structural Change 4:

The 2023 Surgical Coding Overhaul

year_of_diagnosis	**rx_summ_surgprim_site19982022**	**rx_summ_surgprim_site202320 23**
2013	populated	
…	populated	
2022	populated	
2023		populated

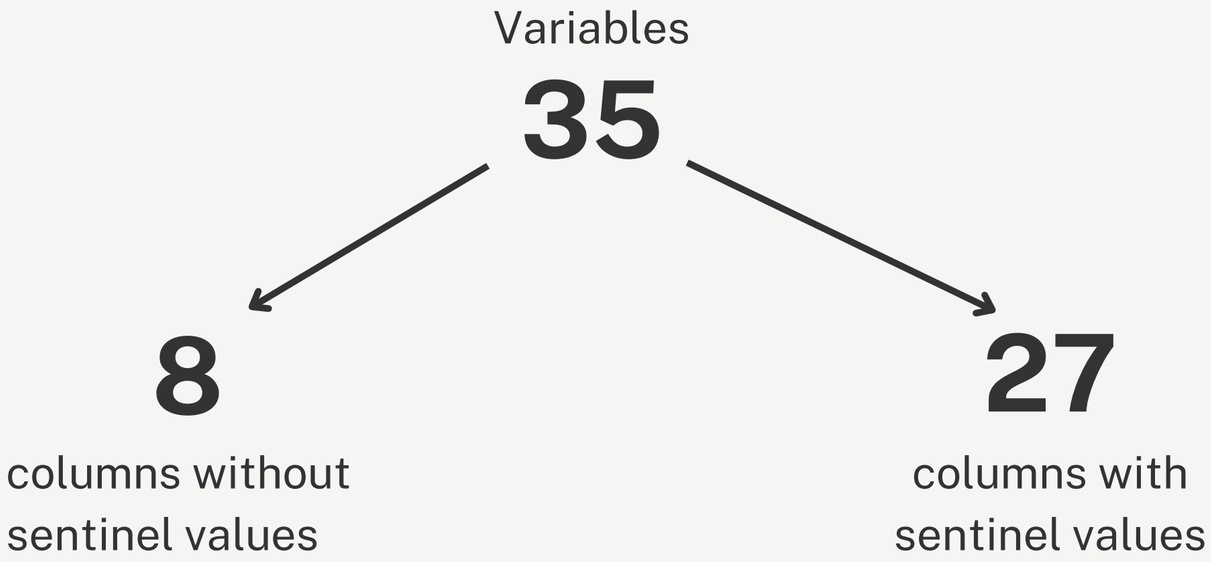
Sentinel Value Study:

In clinical surveillance registries (like NCI SEER and NAACCR), missing or irregular clinical data is rarely stored as a standard NULL or generic NaN.

Instead, registries record sentinel values: domain-specific numeric placeholders (e.g., 999, 98) or standardized strings (e.g., 'Blank(s)', 'Unknown') that explicitly document why a clinical measurement or procedure was not recorded in the patient's chart.

Usually through structural changes, we do see columns becoming systematically blank or a patient being inelligible for lets say, radiation therapy. These systematic gaps/implausibility are shown through sentinel values

Sentinel Value Study



Sentinel Value Study

complete list in the appendix section!

Col #	Raw Column Name	NAACCR Item #	Exact Literal Cell Values / Strings in Dataset	True Clinical / Registry Meaning in SEER	Correct Feature Pipeline Action
3	`age_recode`	SEER Recode	`'90+ years'` ($N = 957$)	Top-coded age bracket mandated by HIPAA Safe Harbor (§164.514)	Extract integer digits $\rightarrow 90.0$; flag `is_age_topcoded=1`
4	`race`	#160	`'Unknown'` ($N = 69$)	NAACCR Code `99`: Unrecorded racial ancestry	Retain as categorical level `'Unknown'`; flag `is_race_unknown=1`
7	`tumor_size_overtime`	SEER Recode	• `'000 (no evidence of primary tumor)'` ($N = 101$) • `'990 (microscopic focus)'` ($N = 8$) • `'998 (site-specific code)'` ($N = 24$) • `'Unknown or size unreasonable (includes any tumor sizes 401-989)'` ($N = 5,134$) • `'Tumor Size Not Consistent Over Time or Not Applicable for this Site'` ($N = 149$)	• **000**: T_0 / Occult primary without macroscopic lung mass • **990**: Qualitative microscopic focus (no measurable dimension given; staging proxy ≤ 1 mm) • **998**: Diffuse involvement of entire lung ($\rightarrow T_4$) • **Unreasonable**: 401–989 mm or unmeasurable size (400 mm SEER biological ceiling) • **Not Consistent**: Code `XX1` (schema shift / inapplicable histology)	Coerce to numeric; non-numeric strings become `NaN`; set flag `is_tumor_size_unknown=1`
8	`tumor_size_summary`	#756	• `'Blank(s)'` ($N = 7,274$) • `'999'` ($N = 3,420$) • `'998'` ($N = 5$) • `'990'` ($N = 5$) • `'988'` ($N = 1$) • `'980'` ($N = 1$), `'910'` ($N = 1$)	• **Blank(s)**: Pre-2016 CS era structural blank (Item #756 introduced 2016) • **999**: Unknown / unmeasurable size • **998**: Site-specific diffuse code ($\rightarrow T_4$) • **990**: Qualitative microscopic focus • **988, 980, 910**: Biologically implausible raw mm (988 mm, 980 mm, 910 mm;	Replace all values ≥ 900 (and text blanks) with `np.nan`; set `is_tumor_size_summary_missing=1`

Sentinel Value Study

Trap	**Column**	**sentinel code**	**models assumption**	**true meaning**
1 meter tumor	tumor_size_summary tumor_size_overtime	990	990mm tumor size [appx 1 meter tumor]	Cells seen only under microscope; no measurable macroscopic mass.
False negative	regional_nodes_positive	98	98 cancer positive nodes	No nodes were examined
0.98 mortality rate signal	seer_combined_metstatdx(bone/brain/liver/lungs)2010	‘Unknown’	Random missing value/noise	Clinically MNAR, patient entered palliative care before imaging could finish [0.98+ mortality rate]

Exploratory Data Analysis

EDA

Unbalanced class

83% patients Dead

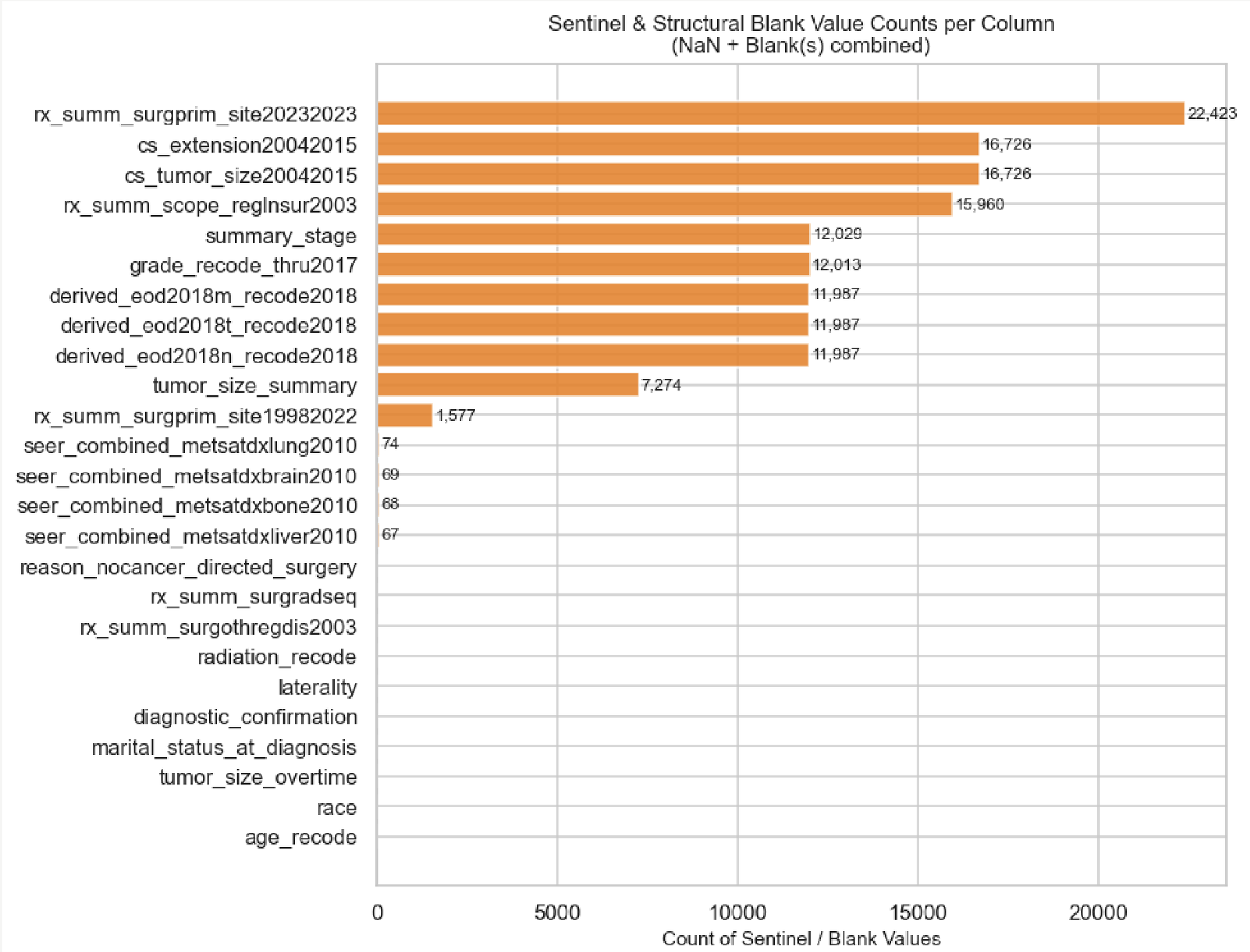
17% patients Alive

EDA

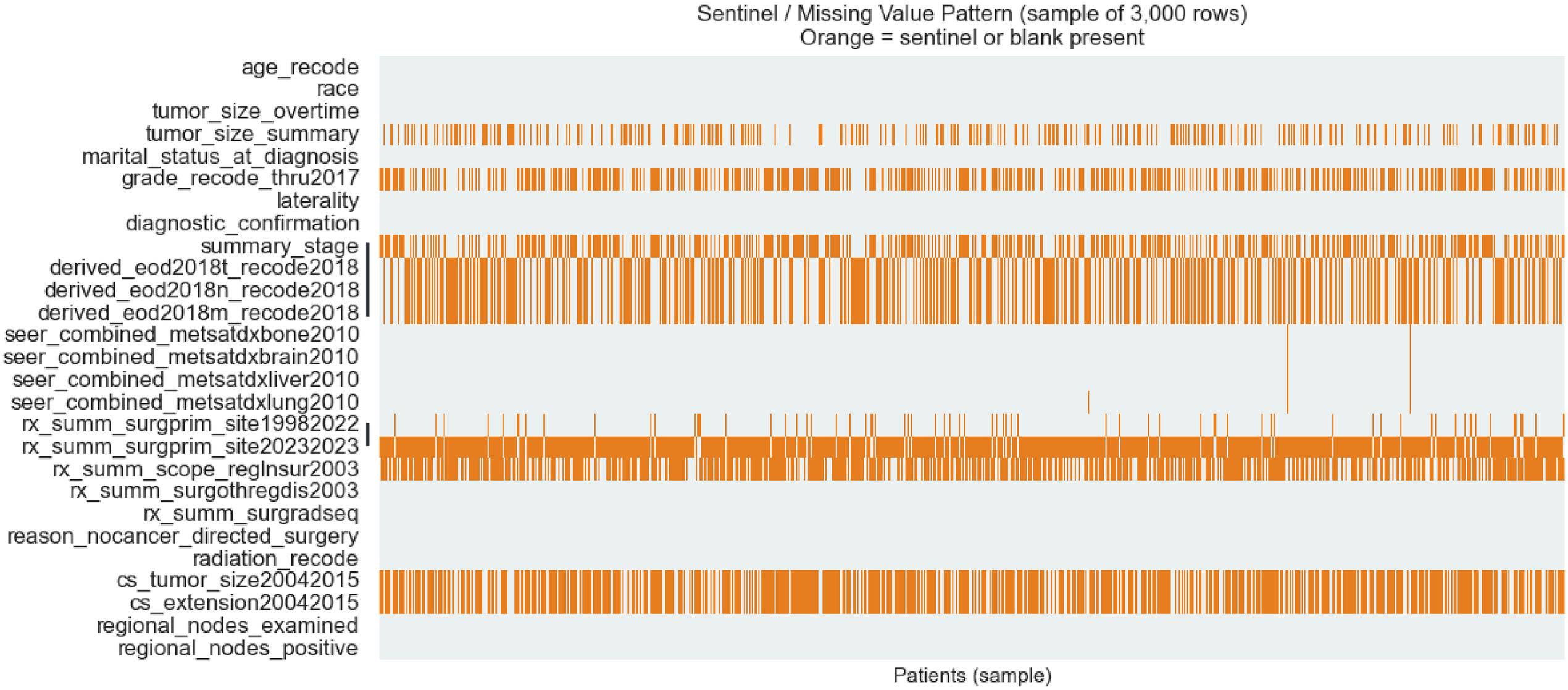
Columns with missing values:

```
{'seer_combined_metsatdxbone2010': 0.3,
 'seer_combined_metsatdxbrain2010': 0.3,
 'seer_combined_metsatdxliver2010': 0.3,
 'seer_combined_metsatdxlung2010': 0.3,
 'rx_summ_scope_reglnsur2003': 66.5}
```

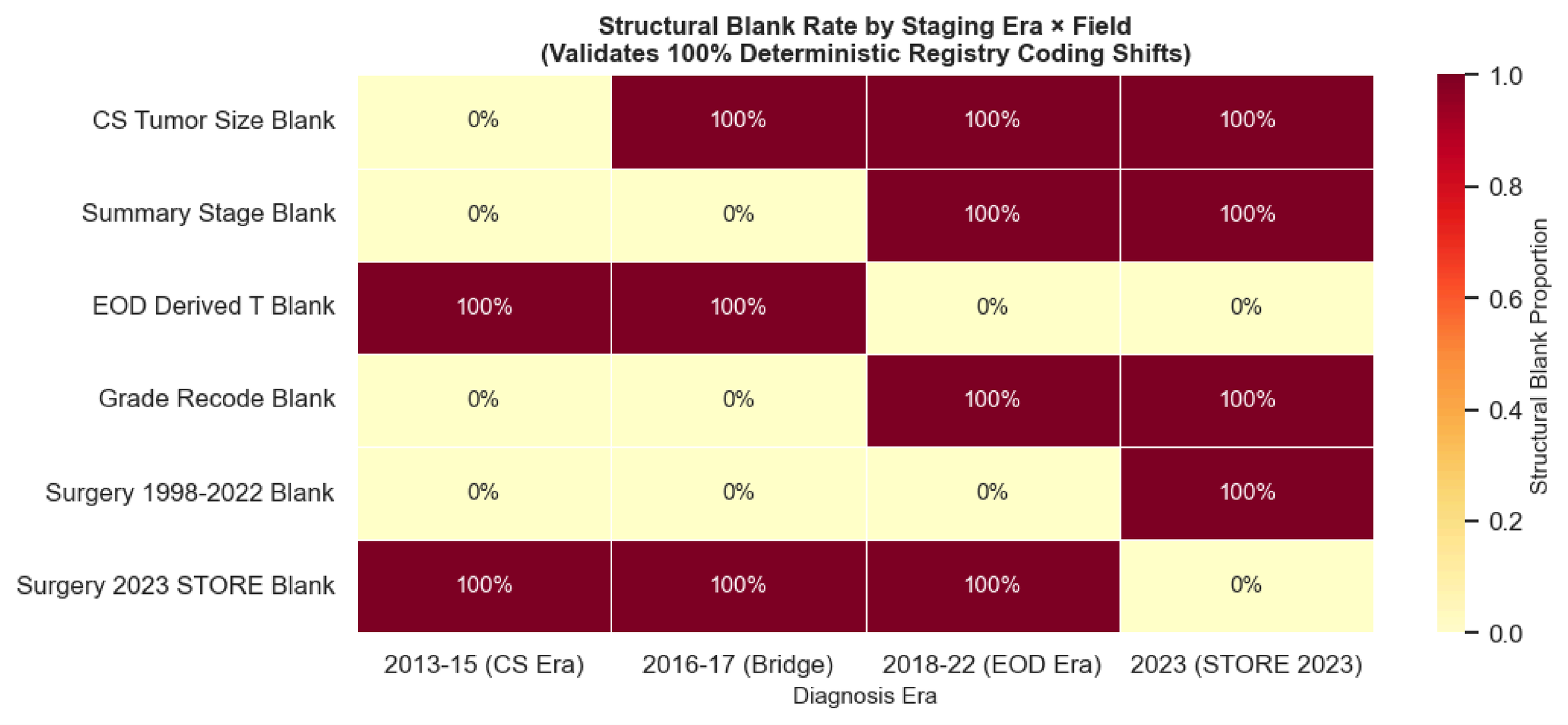
EDA


Sentinel & Structural Blank Value Counts per Column
(NaN + Blank(s) combined)
rx_summ_surgprim_site20232023
cs_extension20042015
cs_tumor_size20042015
rx_summ_scope_reglnsur2003
summary_stage
grade_recode_thru2017
derived_eod2018m_recode2018
derived_eod2018t_recode2018
derived_eod2018n_recode2018
tumor_size_summary
rx_summ_surgprim_site19982022
seer_combined_metsatdxlung2010
seer_combined_metsatdxbrain2010
seer_combined_metsatdxbone2010
seer_combined_metsatdxliver2010
reason_nocancer_directed_surgery
rx_summ_surgradseq
rx_summ_surgothregdis2003
radiation_recode
laterality
diagnostic_confirmation
marital_status_at_diagnosis
tumor_size_overtime
race
age_recode
22,423
16,726
16,726
15,960
12,029
12,013
11,987
11,987
11,987
7,274
1,577
74
69
68
67
0
5000
10000
15000
20000
Count of Sentinel / Blank Values

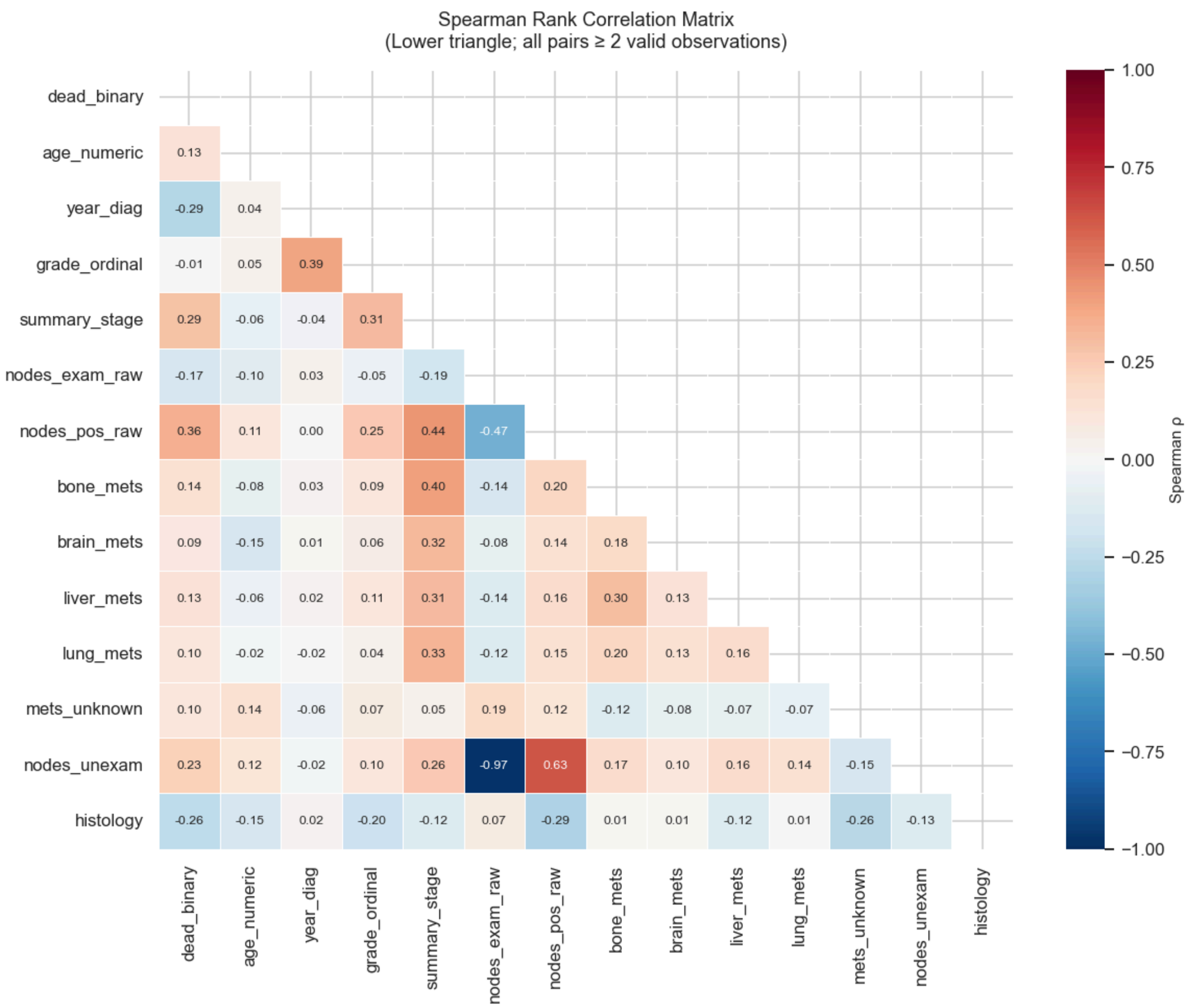
Marked group complements each other



Structural missingness



Rank correlation between features:

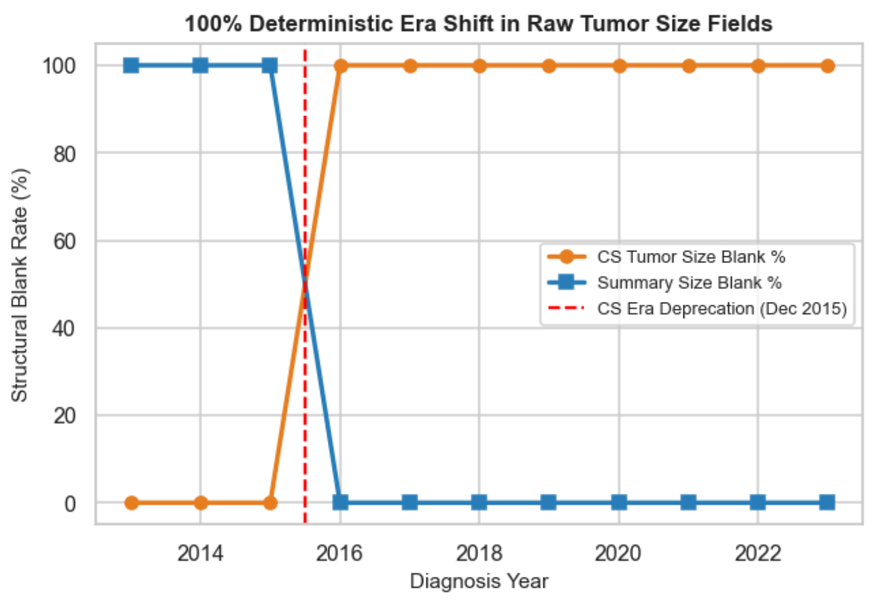


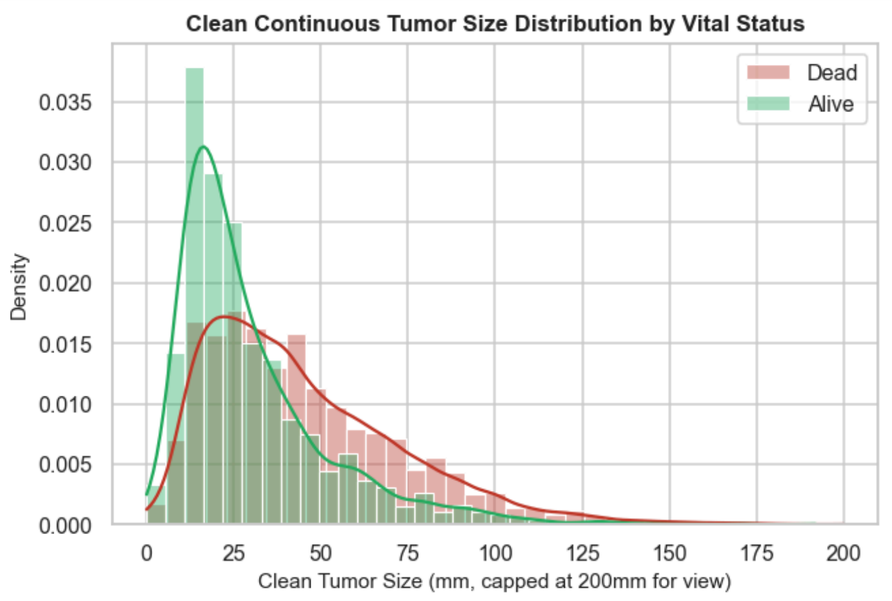
EDA altogether with Data Cleaning & Feature Engineering

Multi-Era Tumor Size Harmonization

```
era_summary = data['tumor_size_summary'].replace('Blank(s)', np.nan)
cs_summary = data['cs_tumor_size20042015'].replace('Blank(s)', np.nan)
combined_era_size = era_summary.fillna(cs_summary)

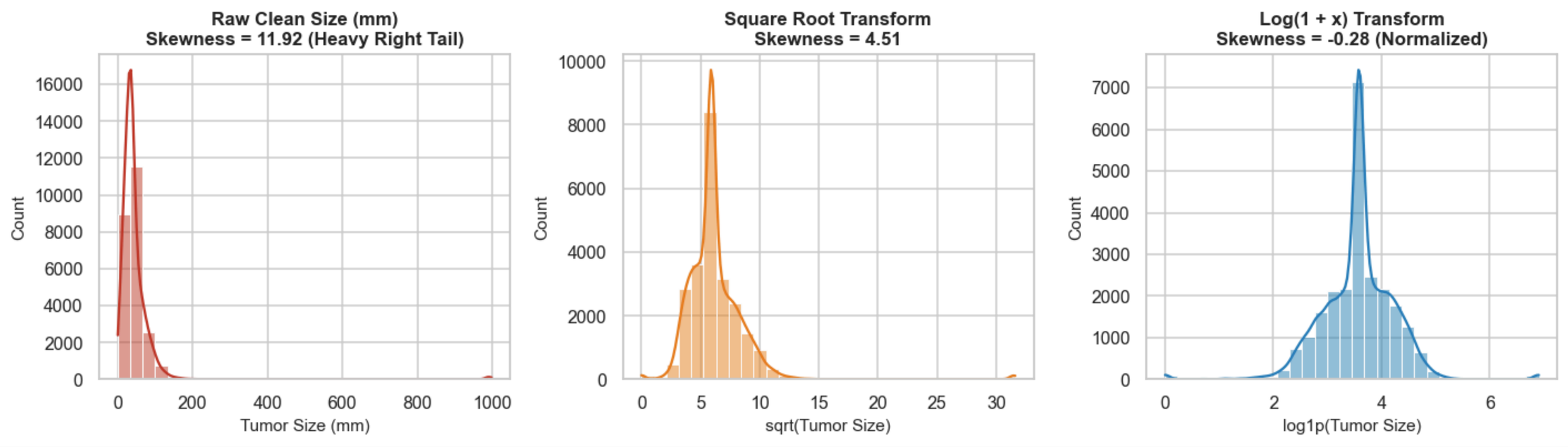
overtime_mm = pd.to_numeric(data['tumor_size_overtime'], errors='coerce')
era_mm = pd.to_numeric(combined_era_size, errors='coerce').replace(999, np.nan)
data['tumor_size_clean_mm'] = overtime_mm.fillna(era_mm)
```





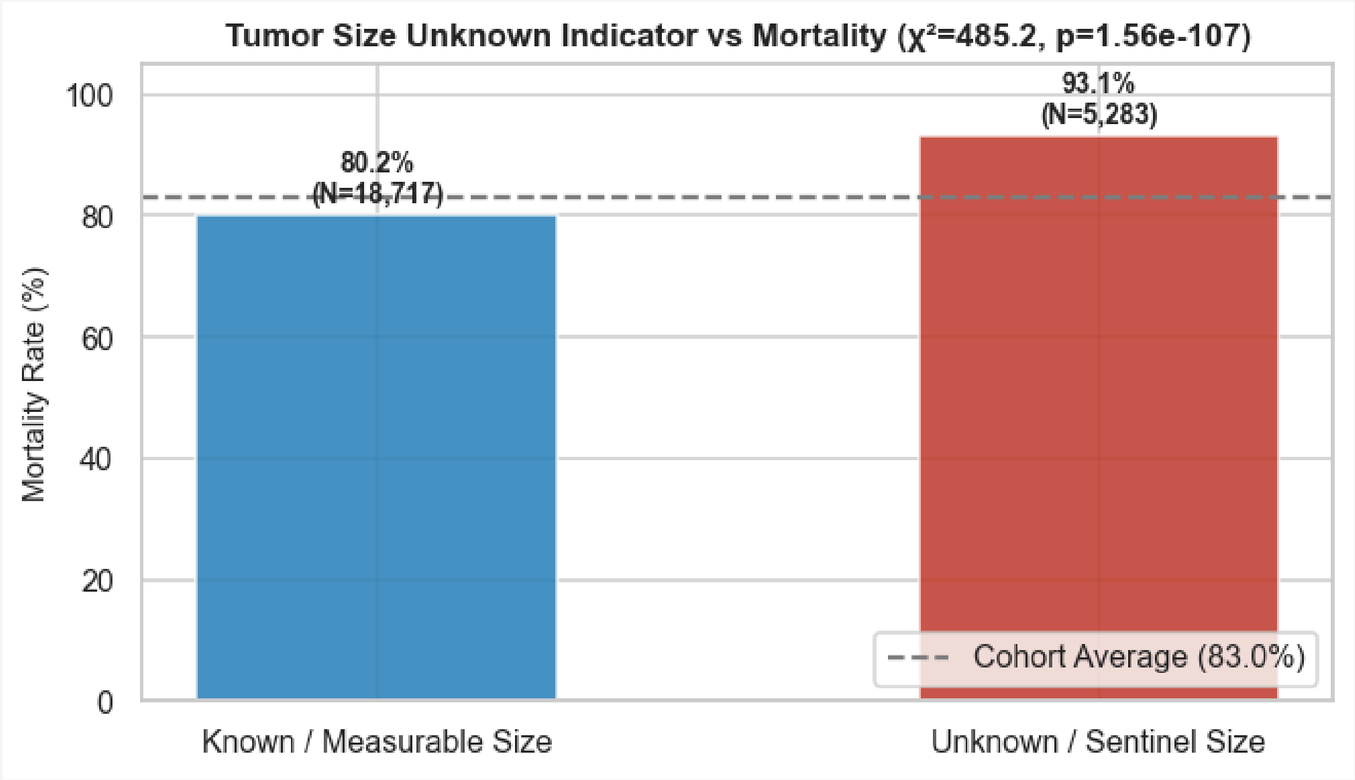
Variance Stabilization (Log & Square Root Size)

```
size_median = data['tumor_size_clean_mm'].median()
imputed_size = data['tumor_size_clean_mm'].fillna(size_median)
data['tumor_size_log'] = np.log1p(imputed_size)
data['tumor_size_sqrt'] = np.sqrt(imputed_size)
```



Unknown Tumor Size Indicator

```
data['is_tumor_size_unknown'] = (
    data['tumor_size_overtime'].astype(str)
    .str.contains('Unknown|unreasonable|Not Consistent', case=False, na=False) |
    data['tumor_size_clean_mm'].isna()
).astype(int)
```



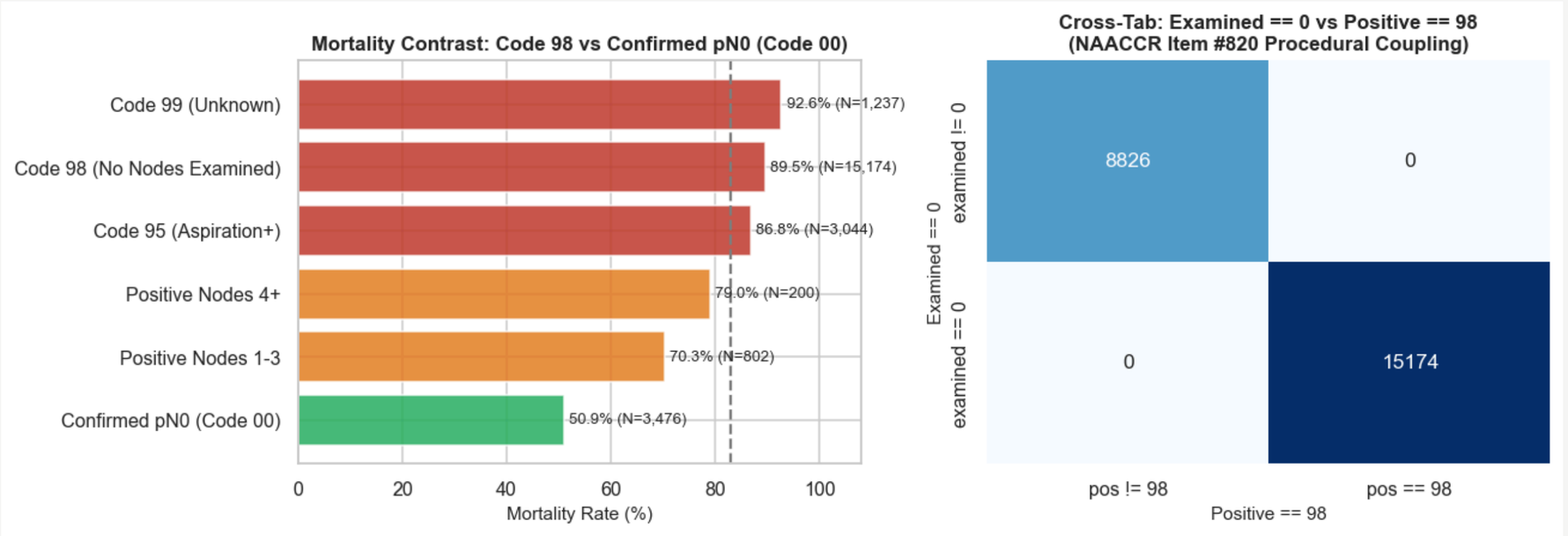
Regional Nodes Code 98 vs 00 & Sentinel Purging

```
exam = pd.to_numeric(data['regional_nodes_examined'], errors='coerce')
pos = pd.to_numeric(data['regional_nodes_positive'], errors='coerce')

# Explicit sentinel indicators
data['nodes_unexamined'] = (pos == 98).astype(int)
data['nodes_exam_unknown'] = (exam == 99).astype(int)

# Purge procedural sentinels >= 90 from continuous counts
pos_clean = pos.copy()
exam_clean = exam.copy()
exam_clean[exam_clean >= 90] = np.nan
pos_clean[pos_clean >= 90] = np.nan

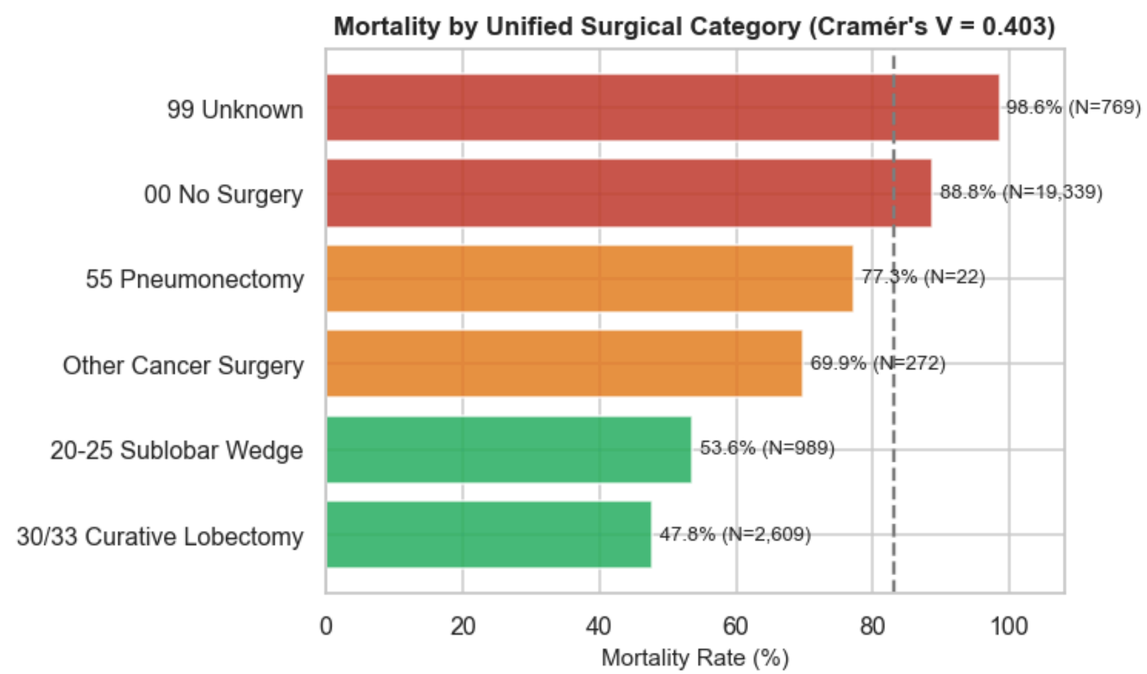
data['nodes_examined_clean'] = exam_clean
data['nodes_positive_clean'] = pos_clean
```

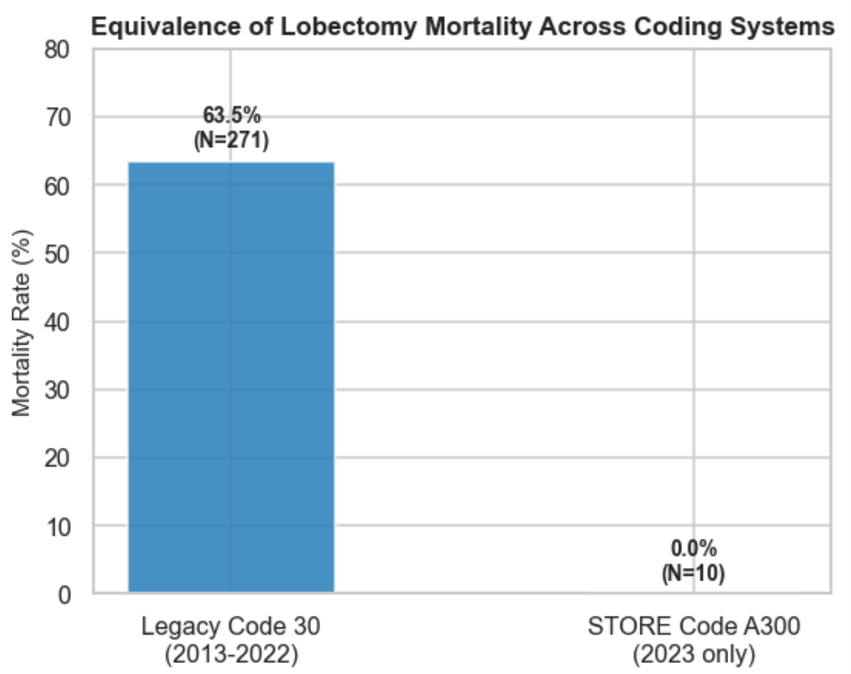


Surgical Code Combining (STORE 2023 vs Legacy 1998–2022)

```python
def unify_surg_code(row):
    v23 = str(row.get('rx_summ_surgprim_site20232023', '')).strip()
    v_old = str(row.get('rx_summ_surgprim_site19982022', '')).strip()
    if v23.startswith('A') and v23 not in ['Blank(s)', 'None', 'nan']:
        digits = v23[1:-1]
        return int(digits) if digits.isdigit() else 99
    return int(v_old) if v_old.isdigit() else 99

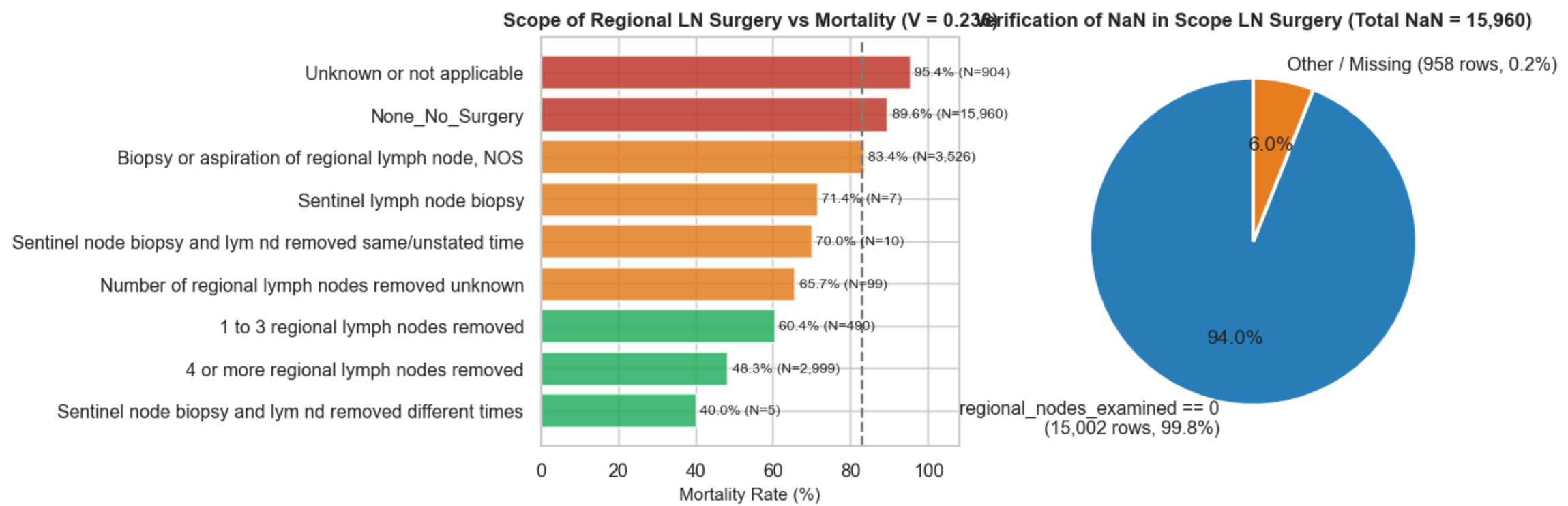
data['unified_surgery_code'] = data.apply(unify_surg_code, axis=1)
```





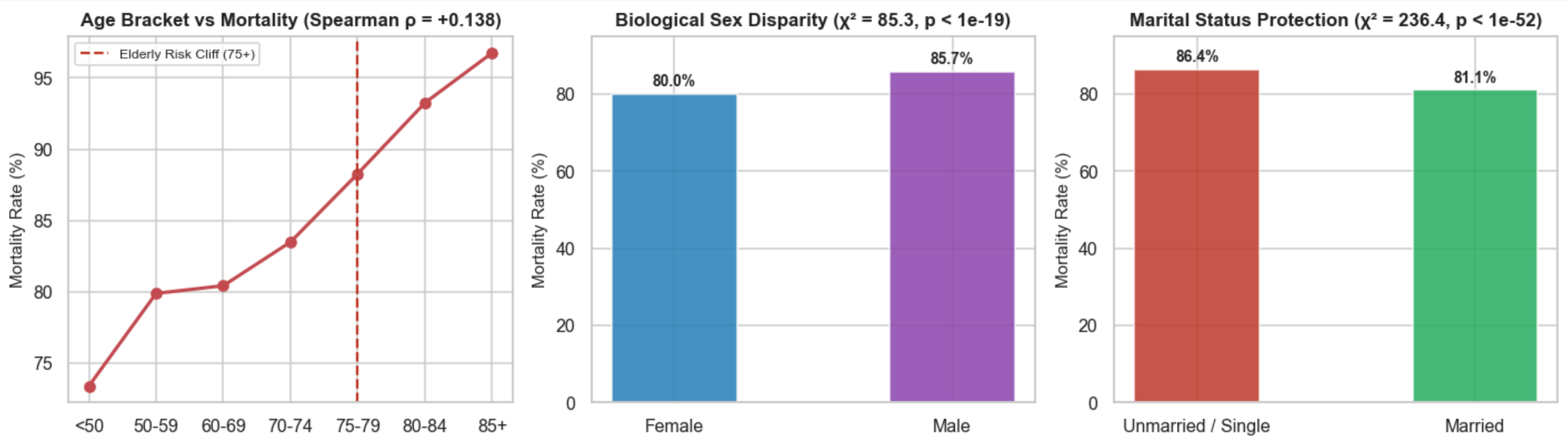
Scope of Regional LN Surgery NaN Imputation

```
data['scope_reg_ln_surgery_clean'] = (
    data['rx_summ_scope_reglnsur2003']
    .replace({'': 'None_No_Surgery', 'Blank(s)': 'None_No_Surgery'})
    .fillna('None_No_Surgery')
)
```



Demographic Characteristics & HIPAA 90+ Top-Coding

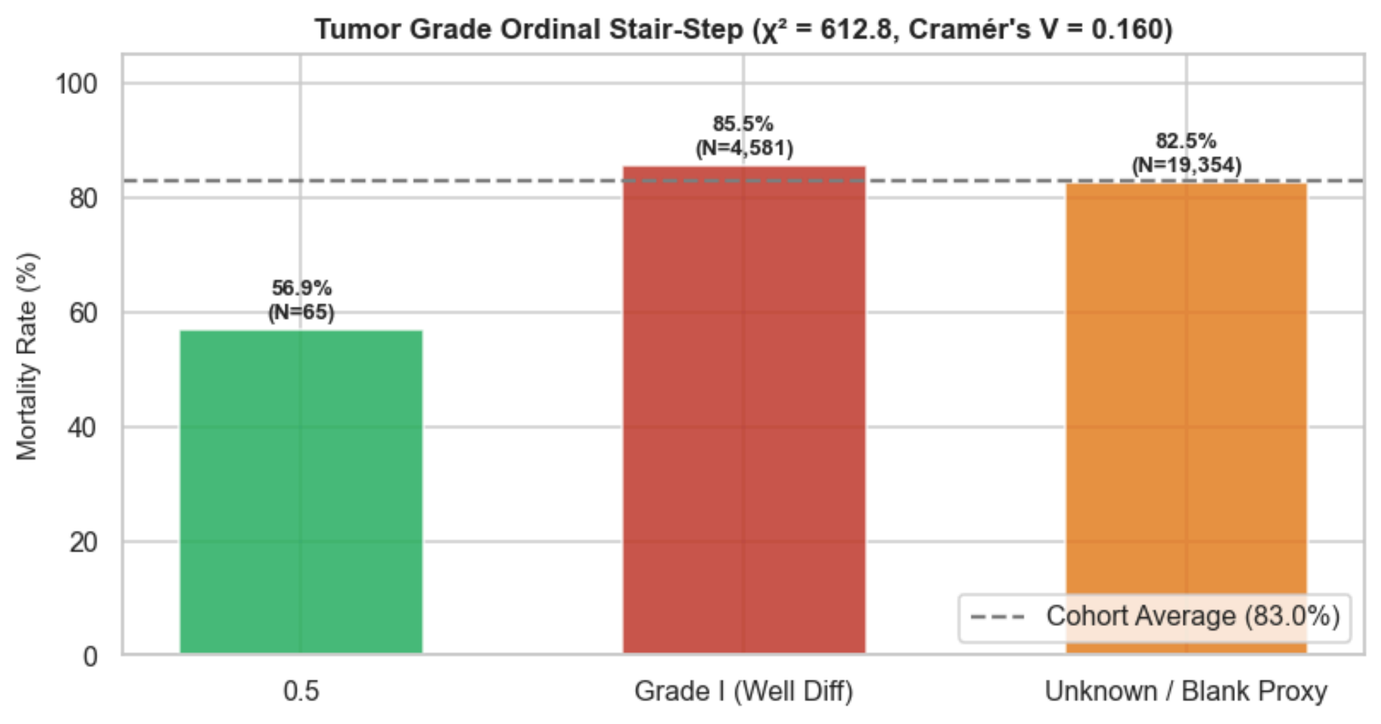
```
data['age_numeric'] = data['age_recode'].astype(str).str.extract(r'(\d+)').astype(float)
data['is_female'] = (data['sex'] == 'Female').astype(int)
data['is_married'] = data['marital_status_at_diagnosis'].astype(str).str.contains('Married', case=False, na=False).astype(int)
data['is_elderly'] = (data['age_numeric'] >= 75).astype(int)
data['is_age_topcoded'] = (data['age_recode'] == '90+ years').astype(int)
data['sequence_numeric'] = data['sequence_number'].astype(str).str.extract(r'(\d+)').astype(float).fillna(1.0)
```



Pathological Differentiation (grade_ordinal & High-Risk Flag)

```python
def parse_grade(val):
    v = str(val).lower()
    if 'grade i' in v or 'well diff' in v: return 1.0
    elif 'grade ii' in v or 'moderately' in v: return 2.0
    elif 'grade iii' in v or 'poorly' in v: return 3.0
    elif 'grade iv' in v or 'undifferentiated' in v or 'anaplastic' in v: return 4.0
    elif 'b-cell' in v or 't-cell' in v: return 0.5
    return 3.5

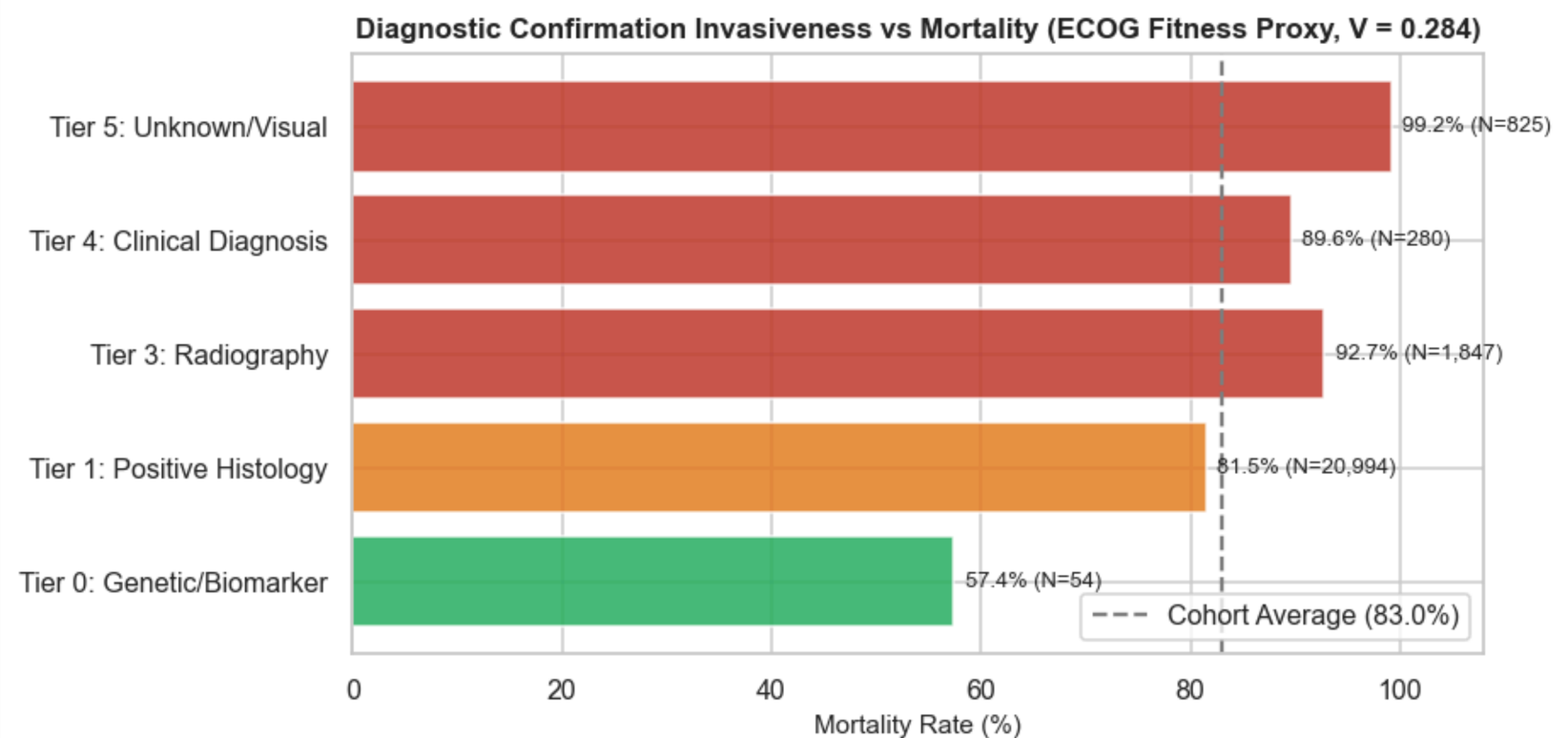
data['grade_ordinal'] = data['grade_recode_thru2017'].apply(parse_grade)
data['is_grade_well_diff'] = (data['grade_ordinal'] == 1.0).astype(int)
data['is_grade_high_risk'] = (data['grade_ordinal'] >= 3.0).astype(int)
```



Diagnostic Invasiveness (diag_confirm_tier & Imaging Flag)

```python
def parse_diag_conf(val):
    v = str(val).lower()
    if 'immunophenotyping' in v or 'genetic' in v or 'laboratory' in v: return 0
    elif 'positive histology' in v: return 1
    elif 'cytology' in v: return 2
    elif 'radiography' in v: return 3
    elif 'clinical diagnosis' in v: return 4
    elif 'unknown' in v or 'visualization' in v: return 5
    return 1

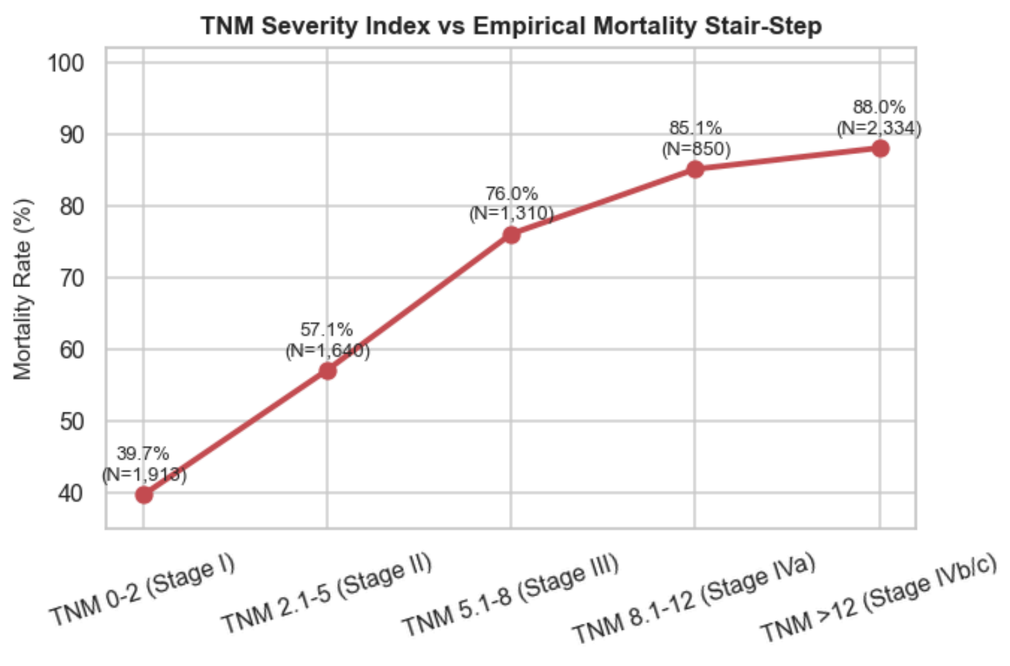
data['diag_confirm_tier'] = data['diagnostic_confirmation'].apply(parse_diag_conf)
data['has_genetic_biomarker_confirm'] = (data['diag_confirm_tier'] == 0).astype(int)
data['is_imaging_only_no_biopsy'] = (data['diag_confirm_tier'] >= 3).astype(int)
```



TNM Severity Index

```
data['eod_t_score'] = data['derived_eod2018t_recode2018'].apply(parse_t_score)
data['eod_n_score'] = data['derived_eod2018n_recode2018'].apply(parse_n_score)
data['eod_m_score'] = data['derived_eod2018m_recode2018'].apply(parse_m_score)

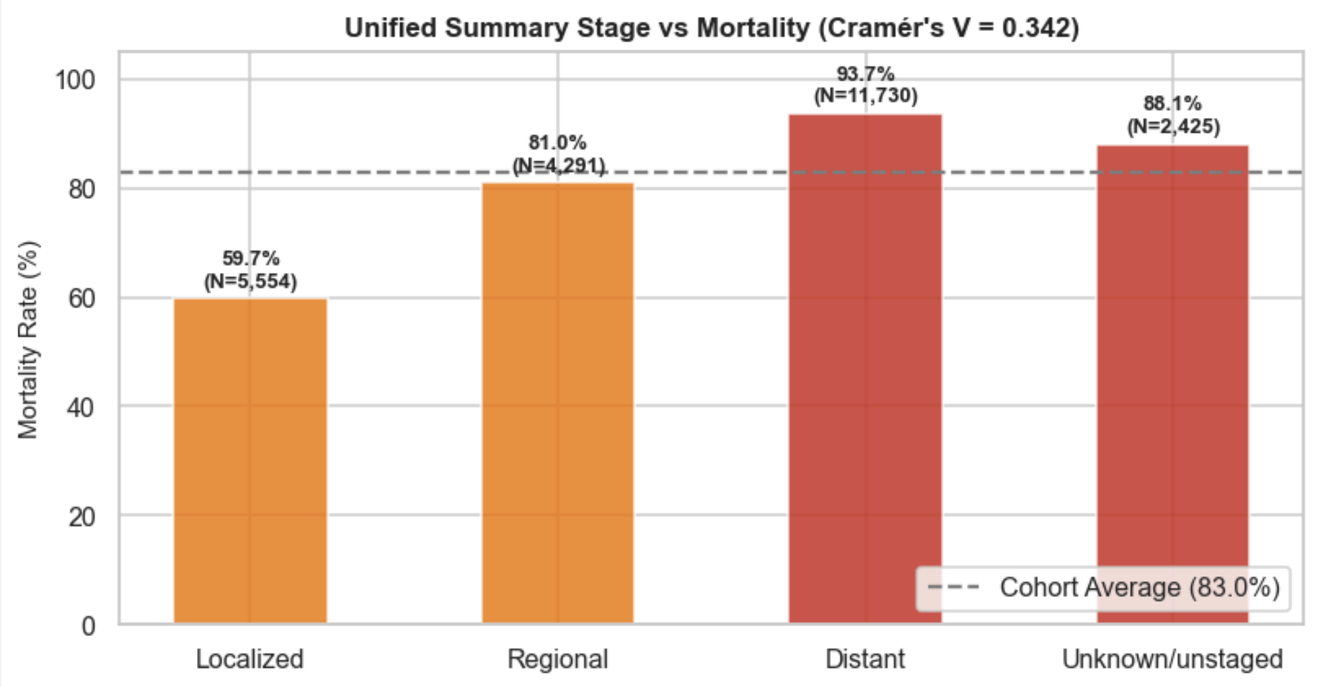
data['tnm_severity_index'] = np.where(
    (data['eod_t_score'] >= 0) & (data['eod_n_score'] >= 0) & (data['eod_m_score'] >= 0),
    data['eod_t_score'] + 2.0 * data['eod_n_score'] + 4.0 * data['eod_m_score'],
    -1.0
)
```



Unified Summary Stage

```python
def map_unified_stage(row):
    stage = str(row.get('summary_stage', '')).strip()
    if stage not in ['Blank(s)', '', 'nan', 'None']: return stage
    m = str(row.get('derived_eod2018m_recode2018', '')).strip()
    n = str(row.get('derived_eod2018n_recode2018', '')).strip()
    t = str(row.get('derived_eod2018t_recode2018', '')).strip()
    if m.startswith('M1'): return 'Distant'
    elif m == 'M0':
        if any(n.startswith(p) for p in ['N1', 'N2', 'N3']): return 'Regional'
        elif n == 'N0' and any(t.startswith(p) for p in ['T1', 'T2', 'T3', 'T4']): return 'Localized'
    return 'Unknown/unstaged'

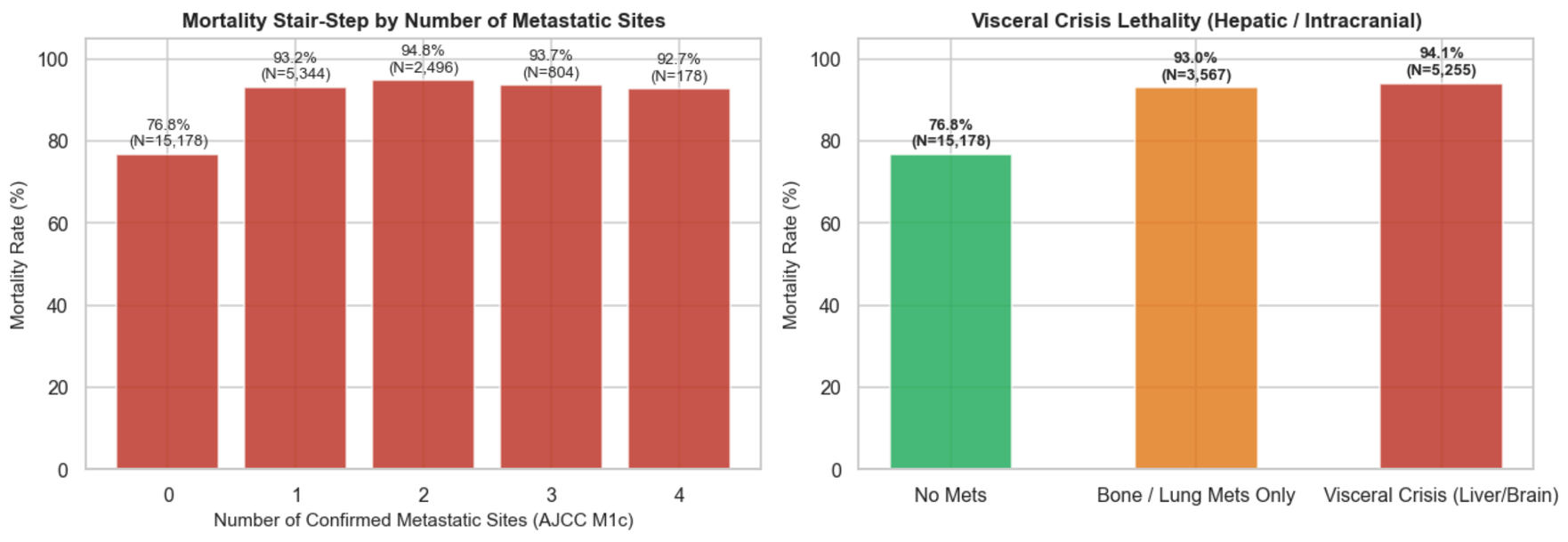
data['unified_summary_stage'] = data.apply(map_unified_stage, axis=1)
stage_map = {'Localized': 1, 'Regional': 2, 'Distant': 3, 'Unknown/unstaged': 0}
data['summary_stage_ordinal'] = data['unified_summary_stage'].map(stage_map).fillna(0).astype(int)
```



Metastatic Multiplicity & Visceral Crisis

```
mets_cols = ['seer_combined_metsatdxbone2010', 'seer_combined_metsatdxbrain2010',
             'seer_combined_metsatdxliver2010', 'seer_combined_metsatdxlung2010']

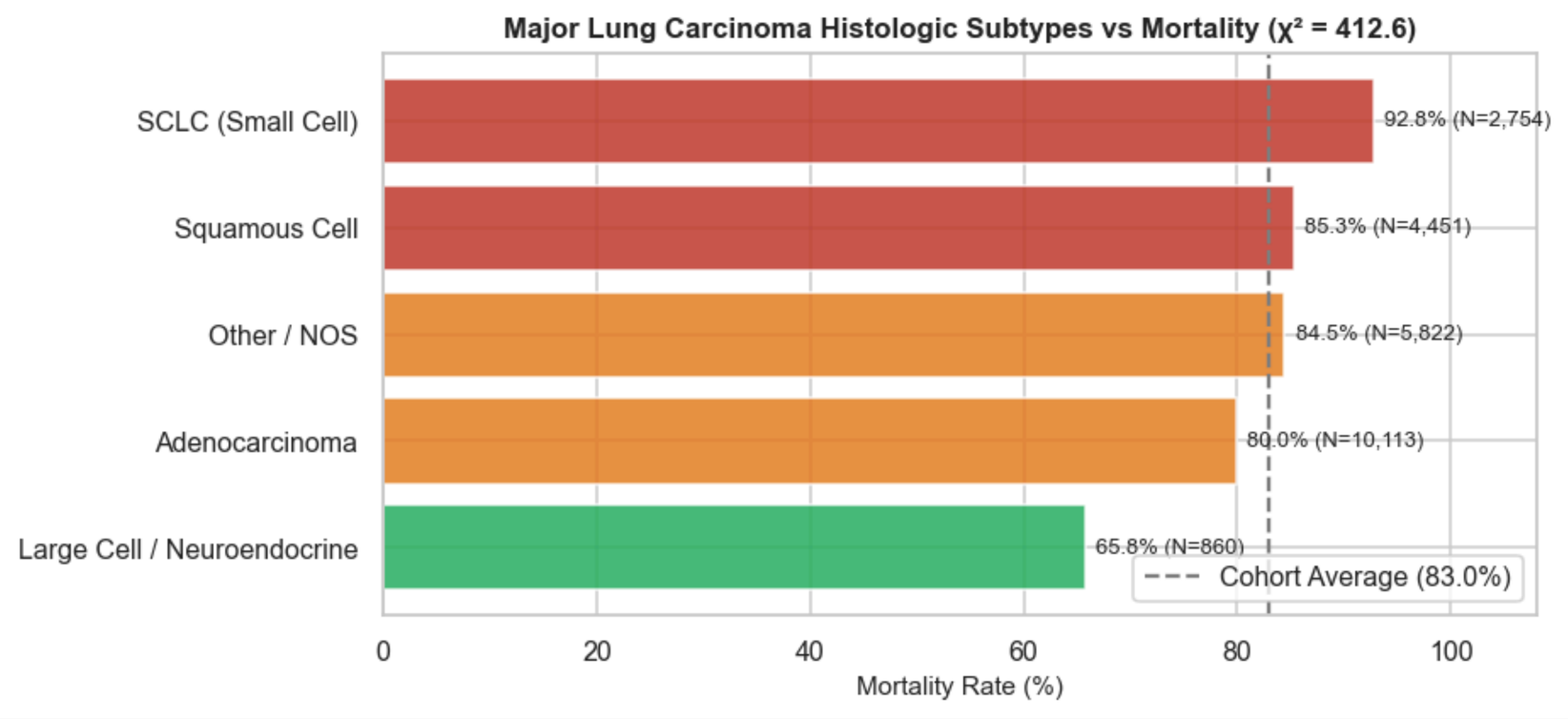
data['metastasis_site_count'] = data[mets_cols].apply(
    lambda row: sum(v == 'Yes' for v in row), axis=1
)
data['has_visceral_crisis'] = (
    (data['has_liver_mets'] == 1) | (data['has_brain_mets'] == 1)
).astype(int)
```



Histology Subtype Risk Hierarchy

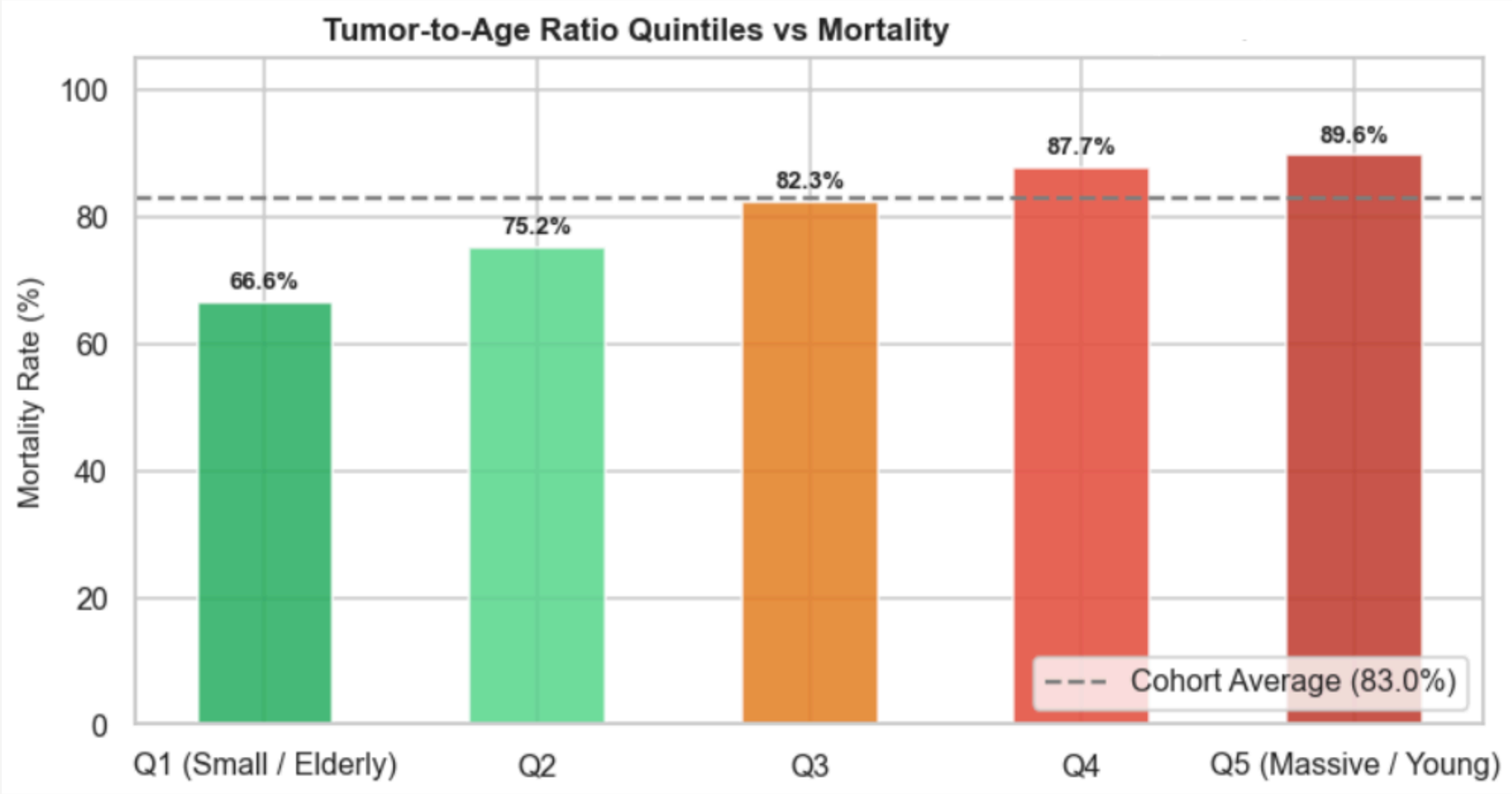
```
hist_num = pd.to_numeric(data['histologic_type_icdo3'], errors='coerce').fillna(-1).astype(int)
def group_histology(code):
    if 8041 <= code <= 8045: return 'SCLC'
    elif code in [8140, 8250, 8251, 8252, 8253, 8254, 8255, 8260, 8480, 8481, 8490, 8550]: return 'Adenocarcinoma'
    elif 8070 <= code <= 8076 or code == 8083: return 'Squamous_Cell'
    elif code in [8012, 8013, 8014, 8020, 8240, 8246, 8249]: return 'Large_Cell_Neuroendocrine'
    return 'Other_NOS'

data['histology_group'] = hist_num.apply(group_histology)
```



Tumor to age ratio vs mortality

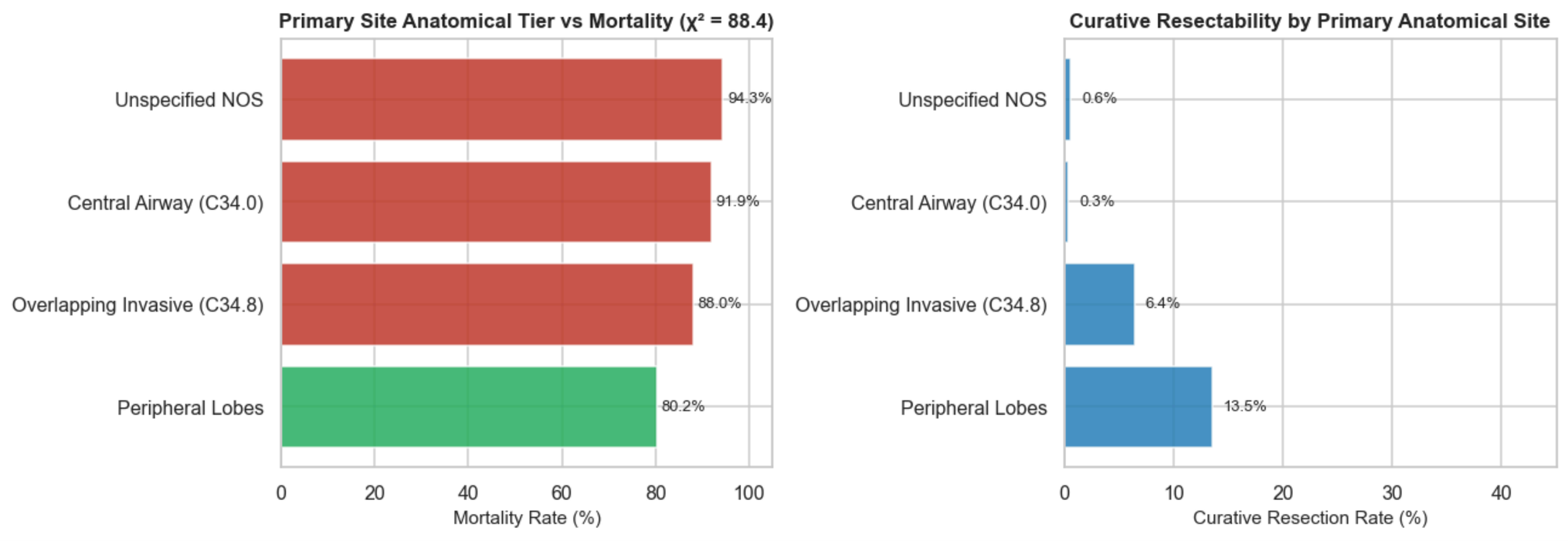
```
data['tumor_to_age_ratio'] = data['tumor_size_clean_mm'] / (data['age_numeric'] + 1.0)
```



Primary Site Location Tier (site_location_tier)

```python
def categorize_site(site_str):
    s = str(site_str).lower()
    if 'main bronchus' in s or 'carina' in s: return 'Central_Airway'
    elif 'upper lobe' in s or 'lower lobe' in s or 'middle lobe' in s: return 'Peripheral_Lobe'
    elif 'overlapping' in s: return 'Overlapping_Invasive'
    return 'Unspecified_NOS'

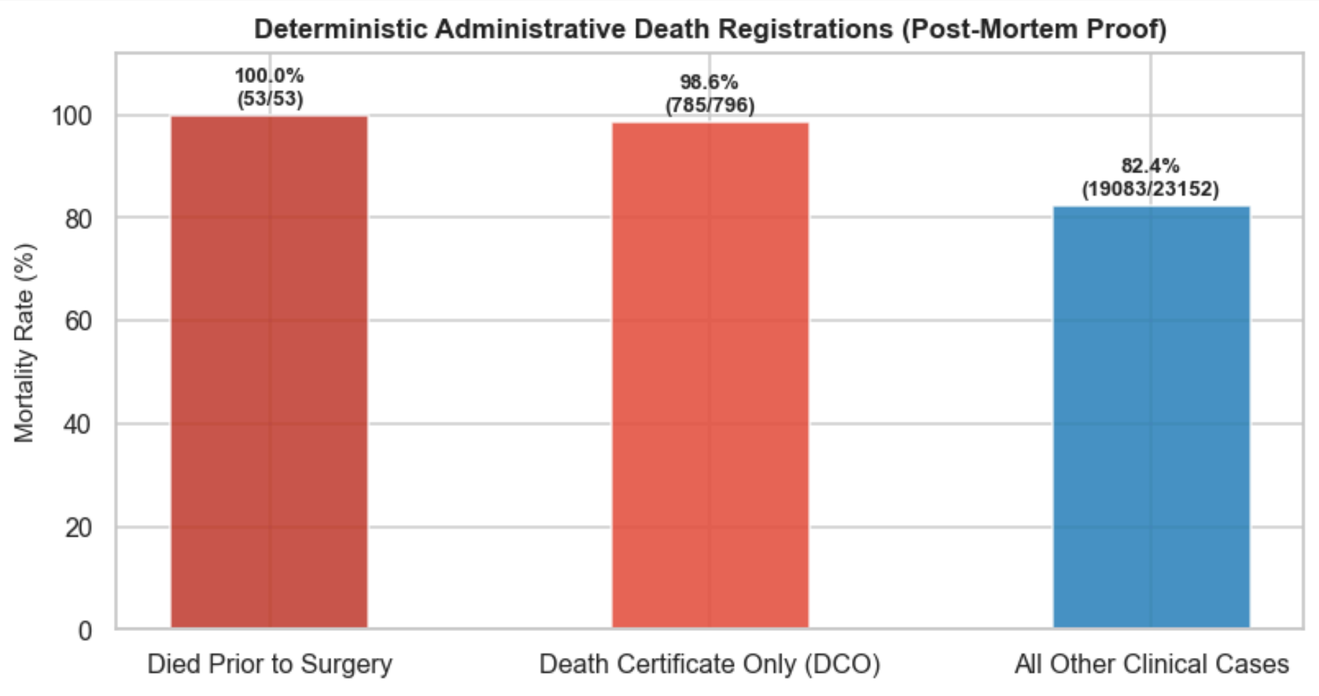
data['site_location_tier'] = data['primary_site'].apply(categorize_site)
data['is_high_risk_site'] = data['primary_site'].astype(str).str.contains('NOS|Main bronchus', case=False, na=False).astype(int)
```



Administrative Survival Overrides

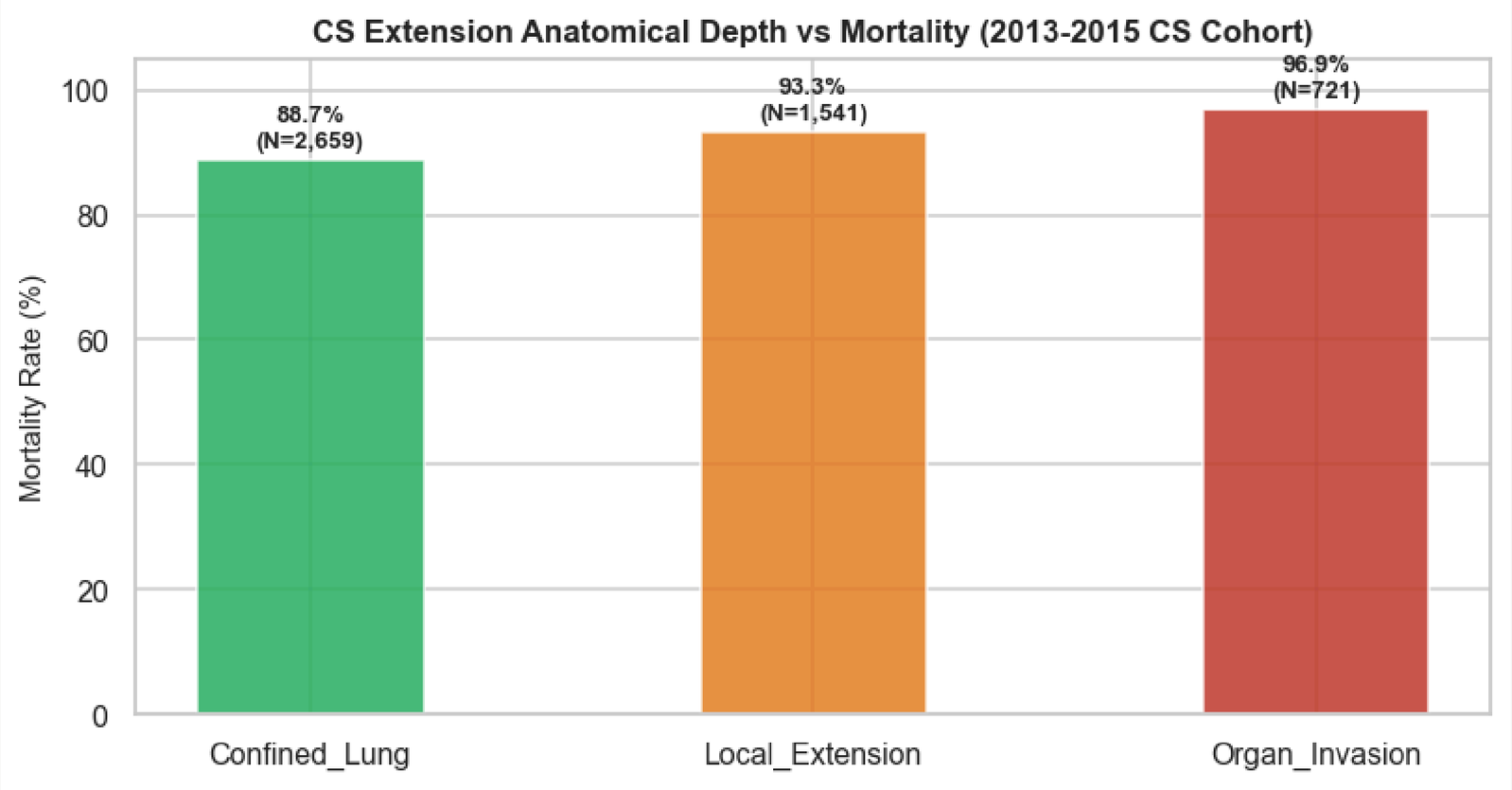
```python
def parse_surg_oth(val):
    v = str(val).lower()
    if 'non-primary' in v and 'distant' not in v: return 0
    elif 'autopsy' in v: return 1
    elif 'distant' in v: return 2
    elif 'death certificate' in v: return 3
    return 1

data['surg_other_tier'] = data['rx_summ_surgothregdis2003'].apply(parse_surg_oth)
data['is_death_cert_only'] = (data['surg_other_tier'] == 3).astype(int)
data['is_extranodal_nhl'] = data['site_recode_icdo3_who2008'].astype(str).str.contains('NHL|Hodgkin', case=False, na=False).astype(int)
```



Collaborative Staging Extension Depth

```
cs_ext = pd.to_numeric(data['cs_extension20042015'], errors='coerce')
data['cs_extension_risk'] = 'Not_CS_Era'
data.loc[cs_ext == 100, 'cs_extension_risk'] = 'Confined_Lung'
data.loc[(cs_ext >= 300) & (cs_ext <= 430), 'cs_extension_risk'] = 'Local_Extension'
data.loc[(cs_ext >= 600) & (cs_ext <= 700), 'cs_extension_risk'] = 'Organ_Invasion'
```



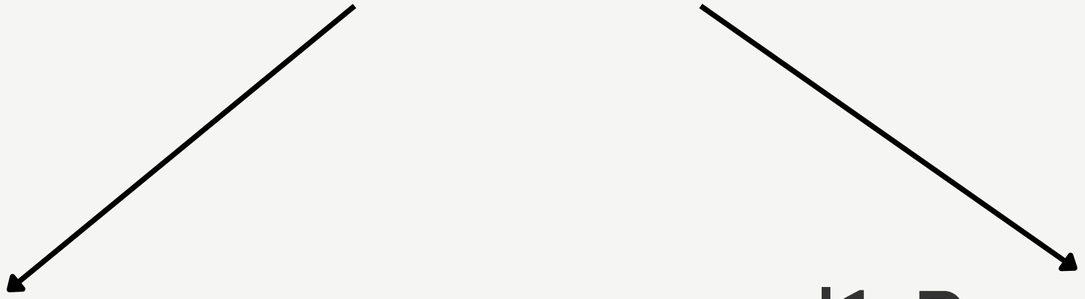
Raw Columns dropped(17):

```
raw_to_drop = [
    'tumor_size_overtime', 'tumor_size_summary', 'cs_tumor_size20042015',
    'regional_nodes_examined', 'regional_nodes_positive',
    'derived_eod2018t_recode2018', 'derived_eod2018n_recode2018', 'derived_eod2018m_recode2018',
    'rx_summ_surgprim_site19982022', 'rx_summ_surgprim_site20232023',
    'rx_summ_scope_reglnsur2003', 'cs_extension20042015', 'summary_stage',
    'grade_recode_thru2017', 'diagnostic_confirmation', 'rx_summ_surgothregdis2003',
    'site_recode_icdo3_who2008'
]
```

Current column number in the train_data: 76

Model Building

Linear Models



Logistic Regression

baseline prediction/inference

l1-Regularized Logistic Regression

Logistic Regression

If the feature engineering were to be applied incase of logistic regression, the design matrix would be zero.

Dropping and keeping the following columns,

```
raw_columns_dropped = ['patient_id', 'race', 'tumor_size_overtime', 'tumor_size_summary', 'origin', 'primary_site',
'site_recode_icdo3_who2008', 'histologic_type_icdo3', 'grade_recode_thru2017', 'laterality', 'diagnostic_confirmation',
'summary_stage', 'rx_summ_surgprim_site19982022', 'rx_summ_surgprim_site20232023', 'rx_summ_scope_reglnsur2003',
'rx_summ_surgothregdis2003', 'rx_summ_surgradseq', 'cs_tumor_size20042015', 'cs_extension20042015']

columns_kept = ['age_numeric', 'is_female', 'is_married', 'is_elderly', 'sequence_numeric', 'tumor_size_log',
'is_tumor_size_unknown', 'grade_ordinal', 'eod_t_score', 'eod_n_score', 'eod_m_score', 'nodes_examined_clean',
'node_positivity_ratio', 'has_bone_mets', 'has_brain_mets', 'has_liver_mets', 'has_lung_mets', 'mets_workup_unknown',
'is_high_risk_site', 'diag_confirm_tier', 'treatment_modality', 'histology_group', 'year_of_diagnosis']
```

Total number of variables(including dummies) in the training set – 37 variables

The following Adjusted Odds Ratio table is given:

	# Coeff (beta) ···	# Std. Error	# p-value	# AOR	# 95% CI Lower	# 95% CI Upper
const	4.507	0.1164	0.0	90.6496	72.1558	113.8834
year_of_diagnosis_2023	-4.6094	0.1452	0.0	0.01	0.0075	0.0132
treatment_modality_Surgery_Only	-2.1507	0.0842	0.0	0.1164	0.0987	0.1373
year_of_diagnosis_2022	-3.3502	0.1421	0.0	0.0351	0.0265	0.0463
year_of_diagnosis_2021	-2.6849	0.1428	0.0	0.0682	0.0516	0.0903
treatment_modality_Radiation_Only	-1.0302	0.0605	0.0	0.3569	0.317	0.4018
year_of_diagnosis_2020	-2.3816	0.1428	0.0	0.0924	0.0698	0.1222
tumor_size_log	0.335	0.0239	0.0	1.3979	1.334	1.4649
year_of_diagnosis_2019	-1.817	0.144	0.0	0.1625	0.1225	0.2155
age_numeric	0.4393	0.0353	0.0	1.5516	1.4477	1.6628
histology_group_Squamous_Cell	0.6399	0.0619	0.0	1.8963	1.6796	2.1408
year_of_diagnosis_2018	-1.5025	0.1461	0.0	0.2226	0.1671	0.2964
treatment_modality_Surgery_and_Radiation	-1.447	0.1444	0.0	0.2353	0.1773	0.3123
has_bone_mets	0.3013	0.0346	0.0	1.3516	1.263	1.4463
is_female	-0.1913	0.0225	0.0	0.8259	0.7903	0.8631

The ANOVA is given

Source	# Resid. Df	# Resid. Deviance	# Deviance Drop (Chi2)	# Df	# Pr(>C...
Null Model (Intercept Only)	23999	21882.54	*Missing value*	*Missing value*	*Missing value*
Full Model (Lean Clinical + Year Dum	23962	13773.16	8109.37	37.0	0.0

- McFadden Pseudo R-squared: 0.3706
  - So the model has strong predictive power.

l1- Regularized Logistic Regression

- Columns retained after the feature engineering and data cleaning- 79 columns = 53 numeric variables + 26 categorical
- After creating dummies for categories = total 171 predictors

Regularization Path Tuning (Minimizing Out-of-Fold Log-Loss):

- Evaluated across a logarithmic grid of regularization strengths ($C \in [0.001, 10.0]$/ $\lambda \in [0.1, 1000]$).
- $C=0.15$ ($\lambda \approx 6.67$) reached the minimum out-of-fold cross-entropy loss, avoiding both over-penalization and variance explosion.

l1- Regularized Logistic Regression

- Total Candidate Predictors: 171
- Retained by Lasso (Non-Zero): 85 (49.7%)
- Pruned to Zero by Lasso: 86 (50.3%)

The ANOVA table given by-

Source	# Resid. Df	# Resid. Deviance	# Deviance Drop (Chi2)	# Df	# Pr(>Chi2) ···
Null Model	23999	21882.54	*Missing value*	*Missing value*	*Missing value*
Lasso Selected Model (85 Active Predictors)	23914	13123.18	8759.36	85.0	0.0

Tri-model Gradient Boosted Decision Trees ensemble model

- Hyper parameter specifications box [static]
  - Catboost: 1,500 iterations,depth=6, learning_rate=0.035, native string category processing.
  - XGboost: 1,500 estimators, max_depth=6, learning_rate=0.030, tree_method='hist'
  - LightGBM: 1,500 estimators, num_leaves=31, max_depth=6, learning_rate=0.030

- Validation Scheme: Strict 5-Fold Stratified Cross-Validation (=24,000, preserving 83% Dead / 17% Alive ratio)

- **Fold-by-Fold Out-of-Fold ROC-AUC Table:**

Fold	CatBoost (GPU)	XGBoost (CUDA)	LightGBM (CPU)
Fold 1	**0.9122**	0.9087	0.9091
Fold 2	**0.9072**	0.9065	0.9066
Fold 3	**0.8946**	0.8924	0.8933
Fold 4	0.8987	**0.8995**	0.8982
Fold 5	0.8913	0.8915	**0.8923**
Mean OOF	**0.9008**	**0.8997**	**0.8999**

Nelder Mead Blending:

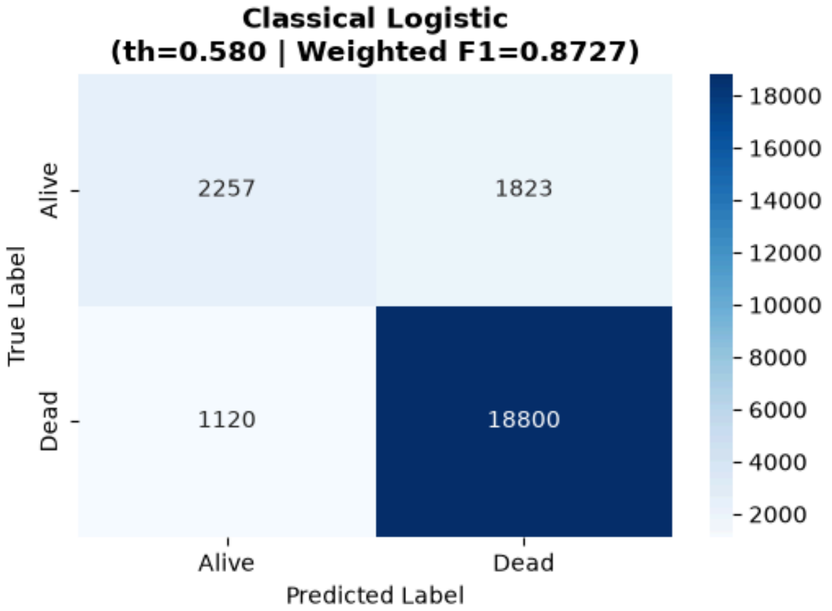
- **Optimization Formula:** $\text{Blend} = w_1 \cdot P_{\text{CatBoost}} + w_2 \cdot P_{\text{XGBoost}} + w_3 \cdot P_{\text{LightGBM}}$
    - Objective: Minimize out-of-fold Log Loss via Nelder-Mead simplex search.
- **Optimal Blend Weights:**
    - **CatBoost (w_1): 0.551 (55.1%)** *(Dominant base learner)*
    - **LightGBM (w_3): 0.325 (32.5%)**
    - **XGBoost (w_2): 0.124 (12.4%)**
    - **Raw Ensemble OOF ROC-AUC:** `0.90161`

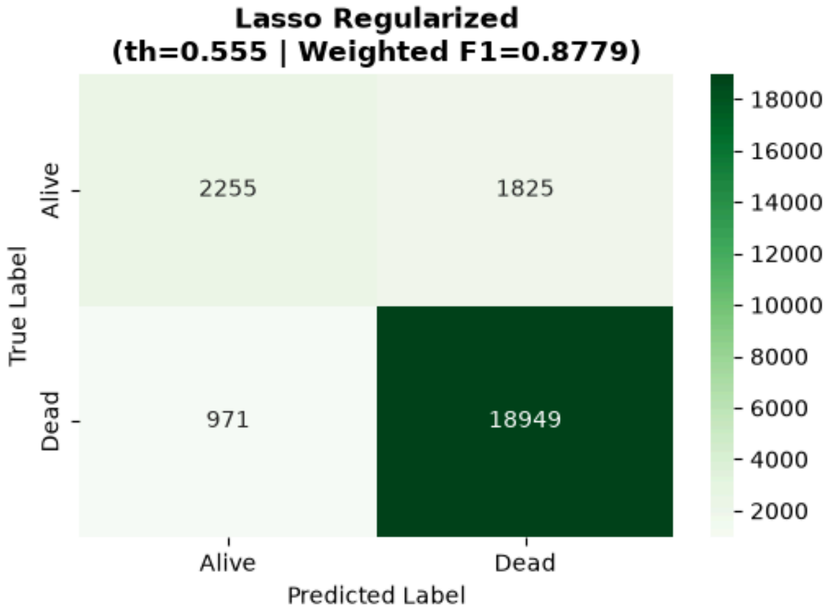
Threshold Tuning (Weighted F1 Optimization)

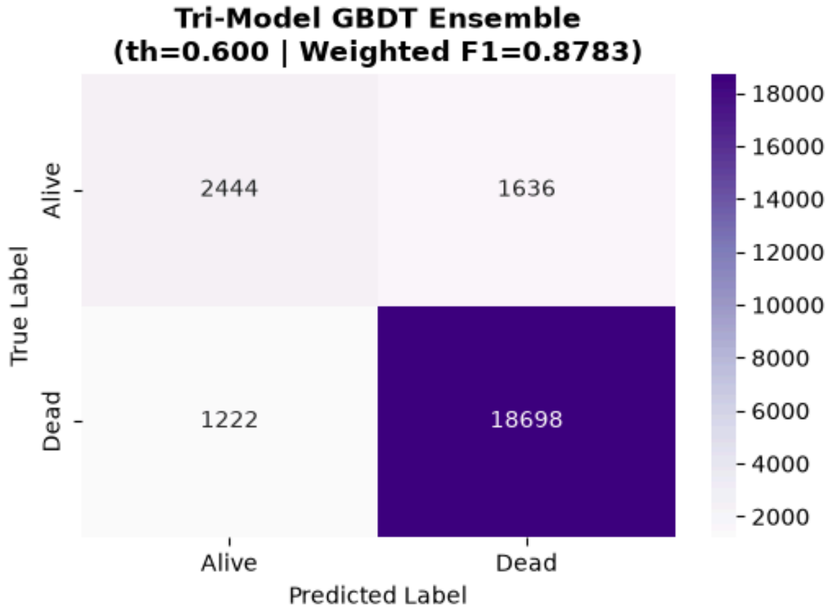
- **Why not 0.50:** With 83% Dead and 17% Alive, default 0.5 favors the majority class and hurts survivor detection.
- **Search approach:** Ran a grid search from **0.40 to 0.70** in **0.005** increments.
- **Selection metric:** Picked the cutoff that maximized **Weighted F1 score**.
- **Evaluation split:**
    - Logistic & Lasso: Tuned on training predictions.
    - Ensemble: Tuned on 5-fold out-of-fold (OOF) predictions.
- **Final thresholds:**
    - **Classical Logistic:** `0.400` (F1: 0.802)
    - **Lasso (L_1):** `0.555` (F1: 0.878)
    - **GBDT Ensemble:** `0.600` (F1: 0.878)

Model Comparison

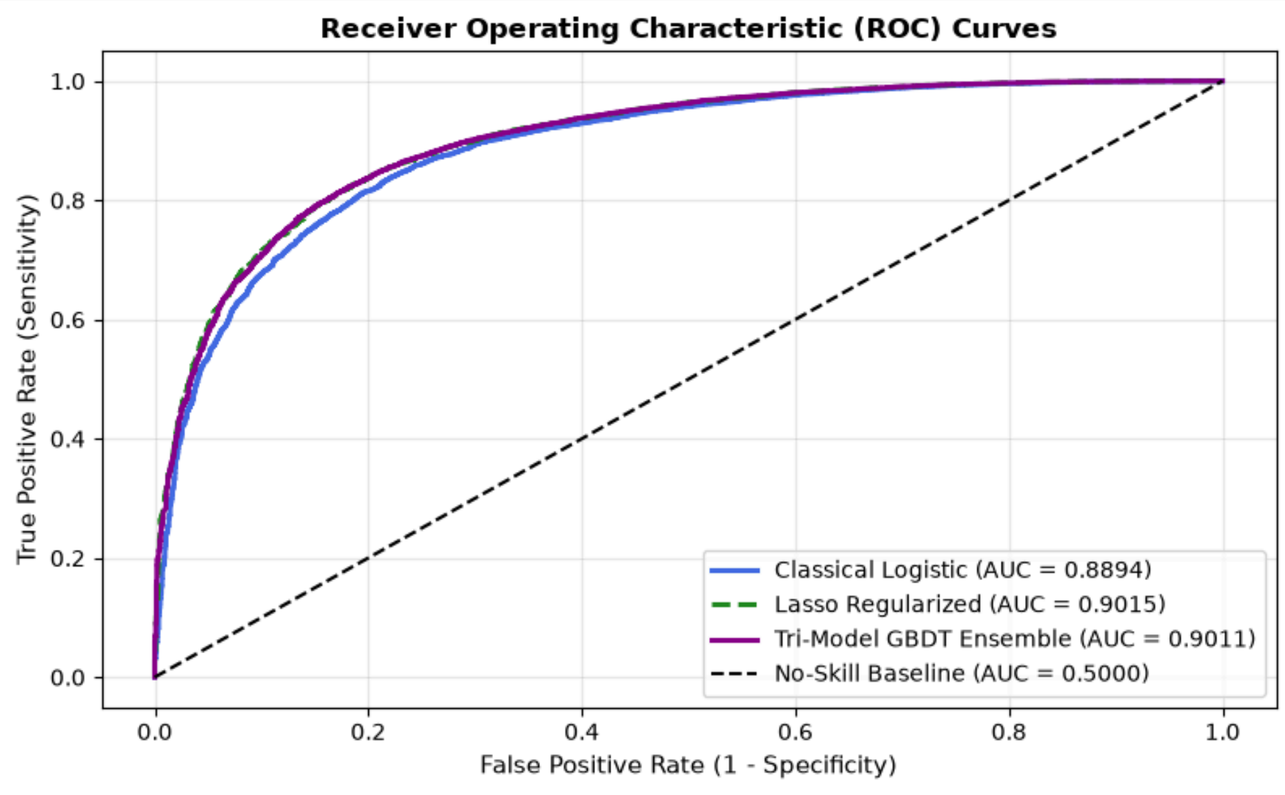
Confusion matrices by each model

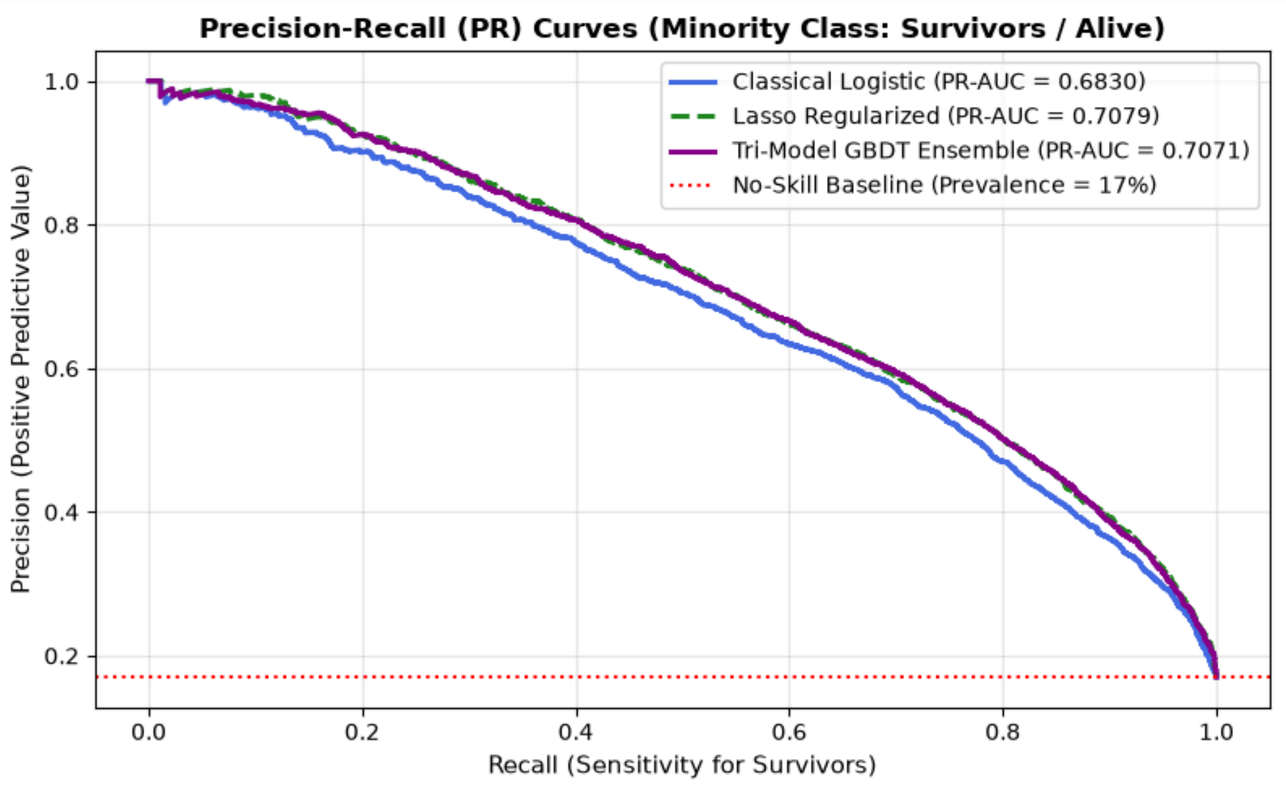






ROC and PR curves





Classification report

```
=== Detailed Classification Report: Classical Logistic (th=0.580) ===
              precision    recall  f1-score   support

   Alive (0)       0.67      0.55      0.61      4080
    Dead (1)       0.91      0.94      0.93     19920

    accuracy                           0.88     24000
   macro avg       0.79      0.75      0.77     24000
weighted avg       0.87      0.88      0.87     24000
```

```
------------------------------------------------------------
=== Detailed Classification Report: Lasso Regularized (th=0.555)
              precision    recall  f1-score   support

   Alive (0)       0.70      0.55      0.62      4080
    Dead (1)       0.91      0.95      0.93     19920

    accuracy                           0.88     24000
   macro avg       0.81      0.75      0.77     24000
weighted avg       0.88      0.88      0.88     24000
```

```
------------------------------------------------------------
=== Detailed Classification Report: Tri-Model GBDT Ensemble (th=0.600)
              precision    recall  f1-score   support

   Alive (0)       0.67      0.60      0.63      4080
    Dead (1)       0.92      0.94      0.93     19920

    accuracy                           0.88     24000
   macro avg       0.79      0.77      0.78     24000
weighted avg       0.88      0.88      0.88     24000
```

Metrics

Model	# Optimal Threshold	# ROC-AUC	# Accuracy (%)	# Sensitivity (%)	# Specificity (%)	# Weighted F1	# Youden J	# Brier ...	# Survivor PR-...
Classical Logistic	0.58	0.8894	87.74	94.38	55.32	0.8727	0.497	0.0871	0.683
Lasso Regularized	0.555	0.9015	88.35	95.13	55.27	0.8779	0.504	0.0834	0.7079
Tri-Model GBDT Ensemble	0.6	0.9011	88.09	93.87	59.9	0.8783	0.5377	0.0834	0.7071

Key findings:

Curative Surgery is the Strongest Protective Factor: Receiving cancer-directed surgery alone reduces the odds of death by **88.4%** ($\mathrm{AOR} = 0.116, p < 0.0001$) compared to untreated disease.

Squamous Cell Carcinoma Carries Higher Hazard: Squamous histology patients have **89.6% higher odds of mortality** ($\mathrm{AOR} = 1.896$) compared to adenocarcinoma.

Female Survival Advantage: Female patients show a **17.4% reduction in mortality odds** ($\mathrm{AOR} = 0.826$) after controlling for age, stage, and treatment.

Temporal Exposure Bias: Diagnosis year 2023 showed an apparent 99% survival benefit ($\mathrm{AOR} = 0.010$). This is **not** a clinical cure—it is **follow-up truncation** (2023 cases had <12 months to reach the mortality endpoint vs. 10 years for 2013 cases).

Any Questions?

References

Standards Body / Authority	Document / Tool Title	Primary URL Reference
NAACCR (North American Association of Central Cancer Registries)	*Data Standards and Data Dictionary, Volume II: Data Item Descriptions (v23 & v24)*	https://datadictionary.naaccr.org/
NCI SEER (National Cancer Institute)	*SEER Program Coding and Staging Manuals (2022–2024)*	https://seer.cancer.gov/tools/codingmanuals/
NCI SEER (National Cancer Institute)	*SEER Extent of Disease (EOD) 2018 Data Standards & Derivation Tools*	https://seer.cancer.gov/eod/
NCI SEER (National Cancer Institute)	*SEER*RSA (Registrar Staging Assistant) — Lung Schema*	https://staging.seer.cancer.gov/
NCI SEER (National Cancer Institute)	*SEER*Stat Variable Definitions & Recode Documentation*	https://seer.cancer.gov/seerstat/variables/seer/
NCI SEER / CDC NPCR	*SEER Summary Stage 2018 (SS2018) & Historical Summary Stage 2000 Manuals*	https://seer.cancer.gov/tools/ss2018/
NCI SEER (National Cancer Institute)	*SEER Cancer Treatment & Radiation Modality Documentation*	https://seer.cancer.gov/data/treatment.html
AJCC (American Joint Committee on Cancer)	*AJCC Cancer Staging Manual (8th Edition) — Lung Chapter & Staging Rules*	https://www.facs.org/quality-programs/cancer-programs/american-joint-committee-on-cancer/
ACS CoC (American College of Surgeons Commission on Cancer)	*STORE 2023 Manual (Standards for Oncology Registry Entry)*	https://www.facs.org/quality-programs/cancer-programs/national-cancer-database/standards-and-data-dictionary/store/
AJCC / NCI SEER	*Collaborative Stage (CSv2) Data Collection System & Lung Schema (2004–2015)*	https://cancerstaging.org/cstage/ & https://seer.cancer.gov/tools/cs/
U.S. HHS OCR (Health & Human Services)	*HIPAA Privacy Rule Safe Harbor Standard for De-identification (45 CFR § 164.514)*	https://www.hhs.gov/hipaa/for-professionals/privacy/special-topics/de-identification/
NCI SEER (National Cancer Institute)	*SEER Hematopoietic and Lymphoid Neoplasm Database & Staging Rules*	https://seer.cancer.gov/tools/heme/